

PORTFOLIO PROJECT

SUBMITTED BY SEJAL DHANTA

PROFESSIONAL BACKGROUND

As a recent graduate with a master's degree in economics and a background in data analytics, I bring a blend of analytical skills, creativity, and adaptability. Throughout my studies and projects, I've honed my abilities in data analysis, especially in areas like exploratory data analysis (EDA), missing data imputation, and metric-based decision-making. My hands-on experience includes projects in car feature analysis, loan default patterns, and hiring process analytics, where I utilized tools like Excel and SQL to draw actionable insights from data.

Additionally, my artistic skills provide a unique perspective, enabling me to create visually engaging and informative dashboards and presentations. I strive to connect data-driven insights to real-world applications, from enhancing operational productivity to improving customer engagement. As a dedicated professional, I am eager to contribute my skills in a data-driven role and make a meaningful impact in organizations that value innovation and detail-oriented analysis.

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1.DATA ANALYTICS IN EVERDAY LIFE

SCENARIO: BUYING A HOME

OVERVIEW:

Data analytics plays a crucial role in buying a home. There are several factors like personal needs and amenities that help in buying a home. Further these factors like planning, preparing, processing, analyzing, sharing and acting upon it help buyers to make informed decisions, optimize their financial investment and makes home buying process simplified.

PLAN:

NEEDS- Willing to buy to lavish 1 storey 3bhk house near Chandigarh, with personal terrace, lawn and parking. Good neighborhood with parks around to play and exercise. Good ambience and amenities. Walking distance to school nearby preferred within a distance of 2.5-3km.

BUDGET- With help of budgeting softwares like mint, personal capital to help evaluate income, savings and current expenditures.

PREPARE:

FINANCE- Willing to spend 1-2.5 crore, money from fixed deposits and savings and willing to mortgage old home and get loan from HDFC bank @8.75% interest rate.

INFORMATION- Gather information about neighbourhood and other properties. Research on housing market trends and local amenities like market park school hospital.

PROCESS:

DATA COLLECTION- Make comprehensive data collection of different properties checking what extra these properties provide as comparative to what all I want.

COMPARISON- Can use statistical approach to compare data and finances included.

ANALYSE:

MARKET TRENDS- Analyse market trends what is best time to buy property and check will property rates increase in upcoming years with advancement in future.

COST ANALYSIS- Conduct total cost analysis that would include insurance, maintenance, amenities, property tax and long-term costs and prepare a budget.

ASTROLOGICAL ANALYSIS- The entrance should be either in the East or North direction. The slope of the house should be in the East or North East as it is considered auspicious. Similarly the rooms, hall, kitchen, bathroom and bedroom should be in a particular direction.

NEIGHBOURHOOD ANALYSIS- Analyse school ratings, crimes rates and history about area.

SHARE:

CONSULT- Present the analysis to financial advisors and family members which will help with facilitating better discussions

HOME VISITS- Schedule appointments to visit some shortlisted properties and making final decisions.

ACT:

MAKE A DECISION- Using tools like decision matrices to check pros and cons of selected option. Final choice should be data driven.

PURCHASE- Negotiate price and complete legal paperwork and buy a home.

Thus, home buying process is simplified.



2.INSTAGRAM USER ANALYSIS

PROJECT DESCRIPTION

- Analyze user interactions and engagement with the Instagram app to provide valuable insights that can help the business grow.
- Tracking how users engage with a digital product, such as a software application or a mobile app.
- Influence the future development of one of the world's most popular social media platforms: INSTAGRAM

APPROACH USED

- For this project I will be using MYSQL to store and analyze database.
- MYSQL is used due to following reasons-
- Easy to understand.
- Allows to access data directly.
- Allows to audit and replicate the data.
- Analyzes multiple tables at once.

MYSQL TASKS

MARKETING ANALYSES

- Loyal User Reward
- Inactive User Engagement
- Contest Winner Declaration
- Hashtag Research
- Ad Campaign Launch

INVESTOR METRICS

- User Engagement
- Bots and Fake accounts

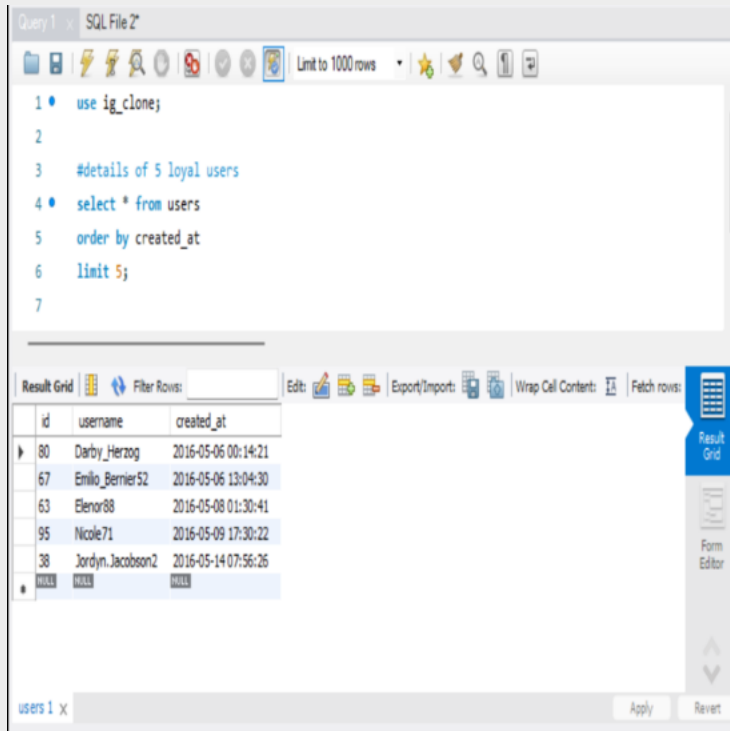
Tech Stack Used

MYSQL WORKBENCH 8.0 CE

MARKETING ANALYSES

TASK: Identify the five oldest users on Instagram from the provided database.

To reward the most loyal users, i.e., those who have been using the platform for the longest time.



The screenshot shows the MySQL Workbench interface. The SQL Editor at the top contains the following query:

```
1 use ig_clone;  
2  
3 #details of 5 loyal users  
4 select * from users  
5 order by created_at  
6 limit 5;  
7
```

Below the editor, the 'Result Grid' tab is active, displaying the results of the query. The results are as follows:

	id	username	created_at
▶	80	Darby_Herzog	2016-05-06 00:14:21
	67	Emilo_Bernier52	2016-05-06 13:04:30
	63	Elenor88	2016-05-08 01:30:41
	95	Nicole71	2016-05-09 17:30:22
	38	Jordyn.Jacobson2	2016-05-14 07:56:26
+	NULL	NULL	NULL

The interface also shows a toolbar with various icons for editing and viewing the results, and a status bar at the bottom indicating 'users 1 x'.

Darby_Herzog, Emilo_Bernier52,

Elenor88, Nicole71 and

Jordyn.Jacobson2

Task: Identify users who have never posted a single photo on Instagram.

Encourage inactive users to start posting by sending them promotional emails.

Query 1 x SQL File 2"

```

6 limit 5;
7
8 • SELECT username
9 from users
10 left join photos
11 on users.id=photos.user_id
12 where photos.id is null;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

username
Aniya_Hackett
Kassandra_Homenick
Jadyn81
Rocio33
Maxwell_Halvorson
Tierra_Trantow
Pearl7
Ollie_Ledner37
McKenna17
David_Osinski47
Morgan_Kassulke
Linnea59
Duane60
Julien_Schmidt
Mike_Auer39
Franco_Keebler64
Nia_Haag
Hulda_Macejkovic
Leslie67
Janelle_Nikolaus81
Darby_Herzog
Esther_Zulauf61
Bartholome_Bernhard
Jessica_West
Esmeralda_Mraz57
Bethany20

Result 2 x | Read Only

A total of 26 Inactive users who haven't posted photos on Instagram

Task: Determine the winner of the contest and provide their details to the team.

The user with the most likes on a single photo wins.

```

53 • Select photos.user_id, users.username, likes.photo_id, count(*) as total_likes
54 from likes likes
55 join photos
56 on photos.id=likes.photo_id
57 inner join users users
58 on photos.user_id=users.id
59 group by likes.photo_id
60 order by total_likes desc
61 limit 1;
62

```

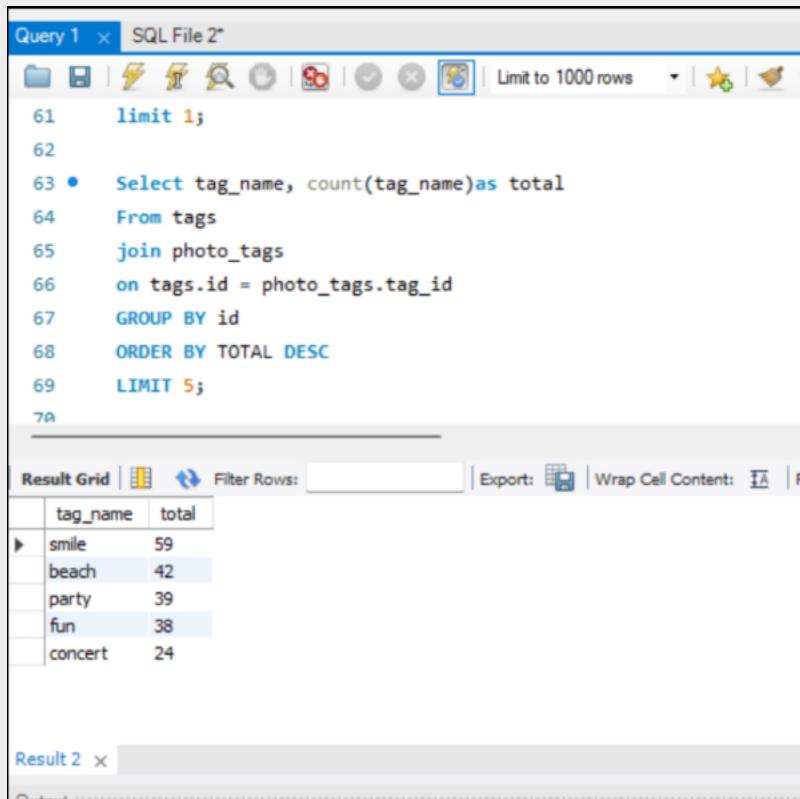
Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows: |

user_id	username	photo_id	total_likes
52	Zack_Kemmer93	145	48

The winner is – Zack_Kemmer93 with 48 likes.

Task: Identify and suggest the top five most commonly used hashtags on the platform.

To know the most popular hashtags to use in their posts to reach the most people.



The screenshot shows a SQL query editor with a query window and a result grid. The query is as follows:

```
61 limit 1;  
62  
63 • Select tag_name, count(tag_name) as total  
64 From tags  
65 join photo_tags  
66 on tags.id = photo_tags.tag_id  
67 GROUP BY id  
68 ORDER BY TOTAL DESC  
69 LIMIT 5;  
70
```

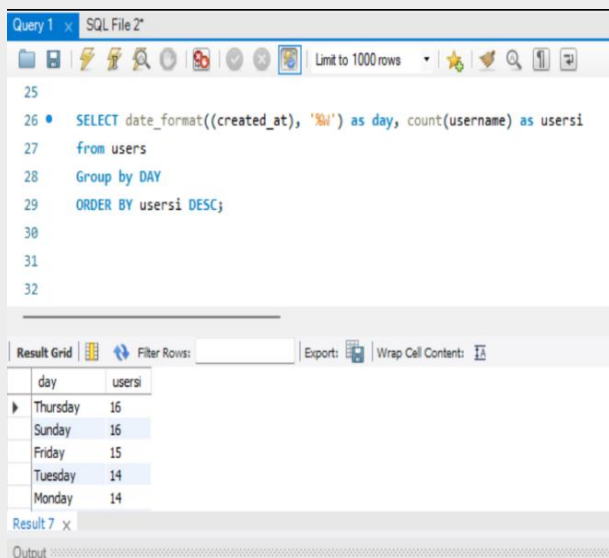
The result grid displays the following data:

tag_name	total
smile	59
beach	42
party	39
fun	38
concert	24

Most used tags are- Smile, beach, party, fun, concert.

Task: Determine the day of the week when most users register on Instagram. Provide insights on when to schedule an ad campaign.

To know the best day of the week to launch ads.



The screenshot shows a SQL query editor with a query window and a result grid. The query is as follows:

```
25  
26 • SELECT date_format((created_at), '%a') as day, count(username) as usersi  
27 from users  
28 Group by DAY  
29 ORDER BY usersi DESC;  
30  
31  
32
```

The result grid displays the following data:

day	usersi
Thursday	16
Sunday	16
Friday	15
Tuesday	14
Monday	14

The best day to launch an ad is THURSDAY

INVESTMENT METRICS

Task: Calculate the average number of posts per user on Instagram. Also, provide the total number of photos on Instagram divided by the total number of users.

To know if users are still active and posting on Instagram or if they are making fewer posts.

```
31 with base as(
32   select u.id as userid, count(p.id) as photoid
33   from users u
34   left join photos p
35   on p.user_id=u.id
36   group by userid)
37   select sum(photoid) as total_photos, count(userid) as total_users,
38   sum(photoid)/count(userid)
39   as photosperuser
40   from base;
```

total_photos	total_users	photosperuser
257	100	2.5700

Average number of posts per user is 2.5700

Task: Identify users (potential bots) who have liked every single photo on the site, as this is not typically possible for a normal user.

To know if the platform is crowded with fake and dummy accounts.

```
71 select users.id, users.username, count(users.id) as total_likes
72 from users
73 join likes
74 on likes.user_id=users.id
75 group by users.id;
```

id	username	total_likes
5	Aniya_Hackett	257
14	JaclynR1	257
21	Rocio33	257
24	Maxwell_Halvorson	257
36	Olle_Ledner37	257
41	Hikerna17	257
54	Duane10	257
57	Julen_Schmidt	257
66	Mike_Auer39	257
71	Nia_Thong	257
75	Leslie7	257
76	Janelle.Nikolaus11	257
91	Bethany20	257

There are 13 fake accounts

RESULT

- Through this project I learned the fundamentals of MYSQL. I came to know how SQL allows easy access of data, is simple and beginner friendly. It is so easy to analyse tables at once and also big chunks of data.
 - Analysing key metrics can help investor make informed decisions and content strategies. Continuous monitoring and adaptation based on data insights will be crucial to maintain and enhance engagement on platform.
-

3.“OPERATION ANALYTICS AND INVESTIGATING METRIC SPIKE”

GOAL: To use your advanced SQL skills to analyze the data and provide valuable insights that can help improve the company's operations and understand sudden changes in key metrics.

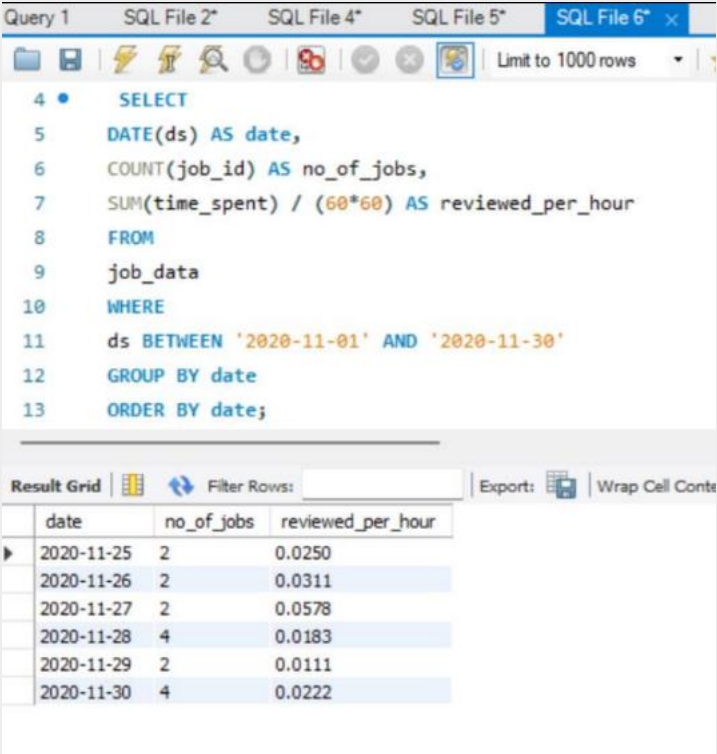
TECH-STACK USED

MYSQL 8.0

CASE STUDY 1: JOB DATA ANALYTICS

Jobs Reviewed Over Time

Objective: Calculate the number of jobs reviewed per hour for each day in November 2020.



The screenshot shows a MySQL SQL editor window with a query and its results. The query is as follows:

```
SELECT
DATE(ds) AS date,
COUNT(job_id) AS no_of_jobs,
SUM(time_spent) / (60*60) AS reviewed_per_hour
FROM
job_data
WHERE
ds BETWEEN '2020-11-01' AND '2020-11-30'
GROUP BY date
ORDER BY date;
```

The results are displayed in a table with the following columns: date, no_of_jobs, and reviewed_per_hour. The data shows the number of jobs reviewed and the average time spent per hour for each day in November 2020.

date	no_of_jobs	reviewed_per_hour
2020-11-25	2	0.0250
2020-11-26	2	0.0311
2020-11-27	2	0.0578
2020-11-28	4	0.0183
2020-11-29	2	0.0111
2020-11-30	4	0.0222

MORE JOBS REVIEWED PER HOUR- 27 NOVEMBER

LESS JOBS REVIEWED PER HOUR- 29 NOVEMBER

Throughput Analysis

Objective: Calculate the 7-day rolling average of throughput (number of events per second).

The screenshot shows a SQL query in a query editor. The query calculates a 7-day rolling average of throughput. The query is as follows:

```
86
87 SELECT
88     ds,
89     COUNT(job_id) AS daily_throughput,
90     AVG(COUNT(job_id)) OVER (
91         ORDER BY ds
92         ROWS BETWEEN 6 PRECEDING AND CURRENT ROW
93     ) AS rolling_7_day_avg
94 FROM
95     job_data
96 GROUP BY
97     ds
98 ORDER BY
99     ds;
100
```

The result grid shows the following data:

ds	daily_throughput	rolling_7_day_avg
2020-11-25	2	2.0000
2020-11-26	2	2.0000
2020-11-27	2	2.0000
2020-11-28	4	2.5000
2020-11-29	2	2.4000
2020-11-30	4	2.6667

PREFERENCE: I generally prefer the 7-day rolling average for throughput analysis, especially when making strategic decisions or identifying trends. The rolling average smooths out the daily volatility, providing a clearer picture of long-term performance. It's particularly helpful in environments where daily operations can be erratic due to various factors like holidays, special events, or unexpected disruptions.

Language Share Analysis

Objective: Calculate the percentage share of each language in the last 30 days.

The screenshot shows a SQL query in a query editor. The query calculates the percentage share of each language in the last 30 days. The query is as follows:

```
16 with lang as
17 (
18     SELECT
19         language, COUNT(language) AS language_count,
20         sum(count(language)) over() as total_language_count
21     FROM
22         job_data
23     WHERE
24         ds BETWEEN '2020-11-01' AND '2020-11-30'
25     GROUP BY language
26 )
27 SELECT
28     language, language_count,
29     (language_count/ total_language_count) * 100 AS percentage_share_per_language
30 FROM lang
31 ORDER BY 2 DESC;
```

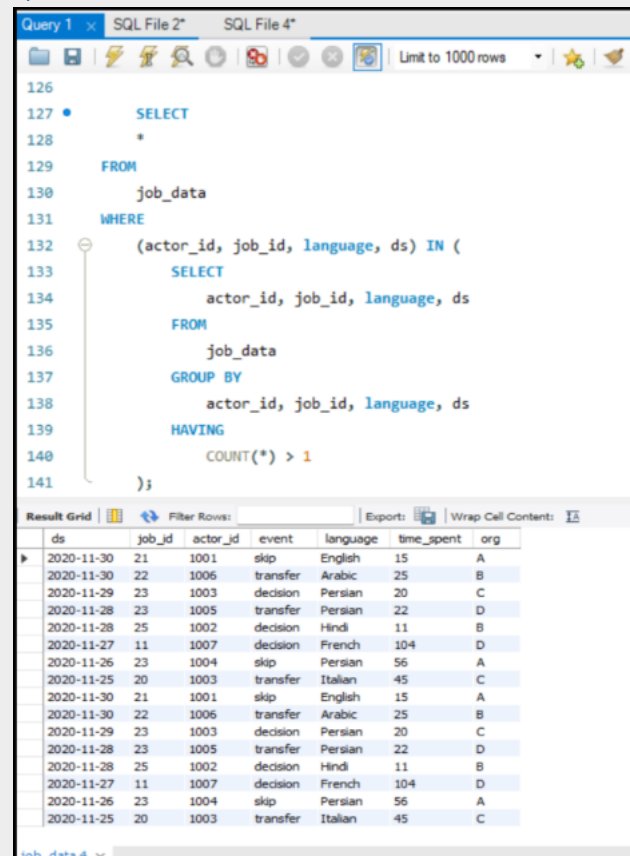
The result grid shows the following data:

Language	language_count	percentage_share_per_language
Persian	6	37.5000
English	2	12.5000
Arabic	2	12.5000
Hindi	2	12.5000
French	2	12.5000

PERSIAN HOLDING HIGHEST SHARE FOLLOWED BY HINDI

Duplicate Rows Detection

Objective: Identify duplicate rows in the data.



```
126
127 • SELECT
128     *
129 FROM
130     job_data
131 WHERE
132     (actor_id, job_id, language, ds) IN (
133         SELECT
134             actor_id, job_id, language, ds
135         FROM
136             job_data
137         GROUP BY
138             actor_id, job_id, language, ds
139         HAVING
140             COUNT(*) > 1
141     );
```

ds	job_id	actor_id	event	language	time_spent	org
2020-11-30	21	1001	skip	English	15	A
2020-11-30	22	1006	transfer	Arabic	25	B
2020-11-29	23	1003	decision	Persian	20	C
2020-11-28	23	1005	transfer	Persian	22	D
2020-11-28	25	1002	decision	Hindi	11	B
2020-11-27	11	1007	decision	French	104	D
2020-11-26	23	1004	skip	Persian	56	A
2020-11-25	20	1003	transfer	Italian	45	C
2020-11-30	21	1001	skip	English	15	A
2020-11-30	22	1006	transfer	Arabic	25	B
2020-11-29	23	1003	decision	Persian	20	C
2020-11-28	23	1005	transfer	Persian	22	D
2020-11-28	25	1002	decision	Hindi	11	B
2020-11-27	11	1007	decision	French	104	D
2020-11-26	23	1004	skip	Persian	56	A
2020-11-25	20	1003	transfer	Italian	45	C

CASE STUDY 2: INVESTIGATING METRIC SPIKE

Weekly User Engagement

Objective: Measure the activeness of users on a weekly basis.

```
3 • SELECT
4   EXTRACT(WEEK FROM occurred_at) AS weeks_of_the_year,
5   COUNT(DISTINCT user_id) AS no_of_users_engaged
6 FROM
7   events
8 WHERE
9   event_type='engagement'
10 GROUP BY weeks_of_the_year
11 ORDER BY 1;
```

weeks_of_the_year	no_of_users_engaged
NULL	6142

User Growth Analysis

Objective: Analyze the growth of users over time for a product.

```
14 • SELECT year, weeknum, num_active_user,
15   SUM(num_active_user) OVER (ORDER BY year, weeknum
16   ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS cum_active_users FROM
17   ( SELECT EXTRACT(year from a.activated_at) AS year,
18     EXTRACT(week from a.activated_at) AS weeknum,
19     COUNT(DISTINCT user_id) AS num_active_user
20   FROM users a GROUP BY year, weeknum ORDER BY year, weeknum) a ;
```

year	weeknum	num_active_user	cum_active_users
NULL	NULL	9381	9381

Weekly Retention Analysis

Objective: Analyze the retention of users on a weekly basis after signing up for a product

```
22 • SELECT
23   EXTRACT(WEEK FROM occurred_at) AS sign_up_week,
24   COUNT(DISTINCT user_id) AS count_of_retention_of_users
25 FROM
26   events WHERE event_type= 'signup_flow '
27   AND event_name= 'complete_signup'
28 GROUP BY sign_up_week
29 order by 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	sign_up_week	count_of_retention_of_users
17	72	
18	163	
19	185	
20	176	
21	183	
22	196	
23	196	
24	229	
25	207	
26	201	
27	222	
28	215	
29	221	
30	238	
31	193	
32	245	
33	261	
34	259	
35	18	

Weekly Engagement Per Device

Objective: Measure the activeness of users on a weekly basis per device.

```
32 • SELECT
33   device,
34   EXTRACT(WEEK FROM occurred_at) AS event_occurance_week,
35   COUNT(DISTINCT user_id) `weekly user engagement per device count`
36 FROM
37   EVENTS
38 WHERE
39   event_type LIKE '%engagements%'
40 GROUP BY device , event_occurance_week
41 ORDER BY 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	device	event_occurance_week	weekly user engagement per device count
	acer aspire desktop	17	9
	acer aspire desktop	18	26
	acer aspire desktop	19	23
	acer aspire desktop	20	23
	acer aspire desktop	21	29
	acer aspire desktop	22	25
	acer aspire desktop	23	22

Email Engagement Analysis

Objective: Analyze how users are engaging with the email service.

```

44 • select
45   (str_to_date(occurred_at,
46     '%d-%m- %Y%H:%i')) as month_name,
47   action,
48   count(user_id) as no_of_users
49   from email_events group by month_name,
50   action
51   order by action desc ;

```

result Grid			Filter Rows:		Export:	
month_name	action	no_of_users				
2014-05-06 09:30:00	sent_weekly_digest	477				
2014-05-13 09:30:00	sent_weekly_digest	491				
2014-05-20 09:30:00	sent_weekly_digest	508				
2014-05-27 09:30:00	sent_weekly_digest	522				
2014-06-03 09:30:00	sent_weekly_digest	538				

4.HIRING PROCESS ANALYTICS

Project Description

This project aims to analyse the hiring process of a large multinational company, like Google, by examining a dataset of past hires. Through this analysis, the objective is to uncover meaningful insights about the company's recruitment trends, including gender distribution, salary analysis, and departmental data. By utilizing Microsoft Excel, this project will provide actionable recommendations to improve the hiring process, streamline recruitment, and optimize employee retention strategies.

To accomplish project objectives (like analysing gender distribution across hires, calculating and evaluating salary patterns, visualizing departmental hiring proportions, examining position tiers and their distributing within the company) a systematic approach is used. This project uses **Microsoft Excel** for data analysis, taking advantage of its versatile functionalities such as formulas, pivot tables, and charting tools. By cleaning and analysing the data, the aim is to present actionable insights that can support better decision-making for the company.

Approach

1. Handling Missing Data: Check if there are any missing values in the dataset. If there are, decide on the best strategy to handle them.
2. Clubbing Columns: If there are columns with multiple categories that can be combined, do so to simplify your analysis.
3. Outlier Detection: Check for outliers in the dataset that may skew your analysis.
4. Removing Outliers: Decide on the best strategy to handle outliers. This could be removing them, replacing them, or leaving them as is, depending on the situation.
5. Data Summary: After cleaning and preparing your data, summarize your findings. This could involve calculating averages, medians, or other statistical measures. It could also involve creating visualizations to better understand the data.

Tech-Stack Used:

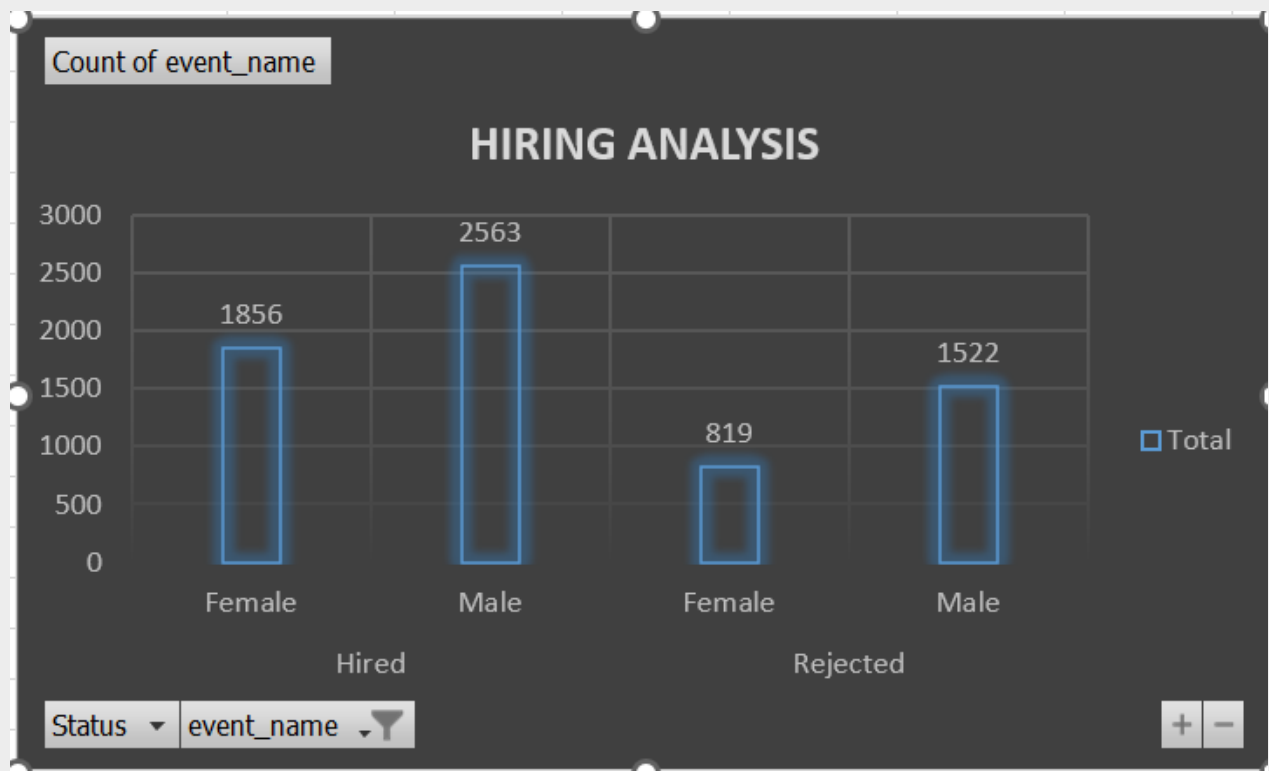
- **Microsoft Excel 2022:** Used for data analysis, formula-based computations, and creating visualizations.

INSIGHTS

Hiring Analysis: The hiring process involves bringing new individuals into the organization for various roles.

Task: Determine the gender distribution of hires. How many males and females have been hired by the company?

Row Labels	Count of event_name
☐ Hired	4419
Female	1856
Male	2563
☐ Rejected	2341
Female	819
Male	1522
Grand Total	6760



Salary Analysis: The average salary is calculated by adding up the salaries of a group of employees and then dividing the total by the number of employees.

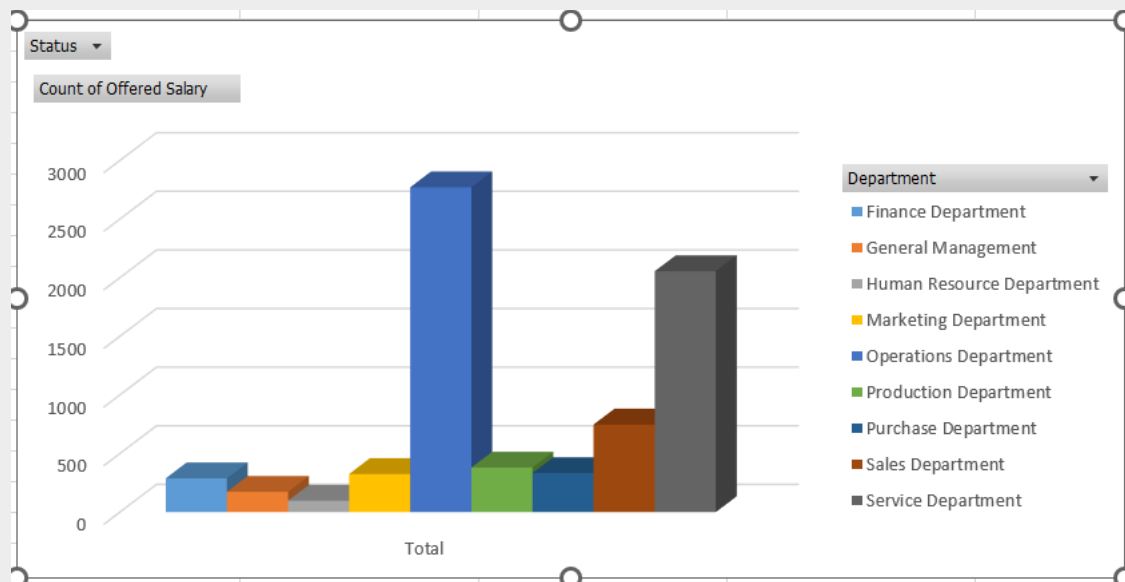
Task: What is the average salary offered by this company? Use Excel functions to calculate this.

Row Labels	Average of Offered Salary	
Finance Department	\$	49,628
General Management	\$	58,722
Human Resource Department	\$	49,002
Marketing Department	\$	48,490
Operations Department	\$	49,151
Production Department	\$	49,448
Purchase Department	\$	52,565
Sales Department	\$	49,310
Service Department	\$	50,630
Grand Total	\$	49,983



Salary Distribution: Class intervals represent ranges of values, in this case, salary ranges. The class interval is the difference between the upper and lower limits of a class.

Task: Create class intervals for the salaries in the company. This will help you understand the salary distribution.

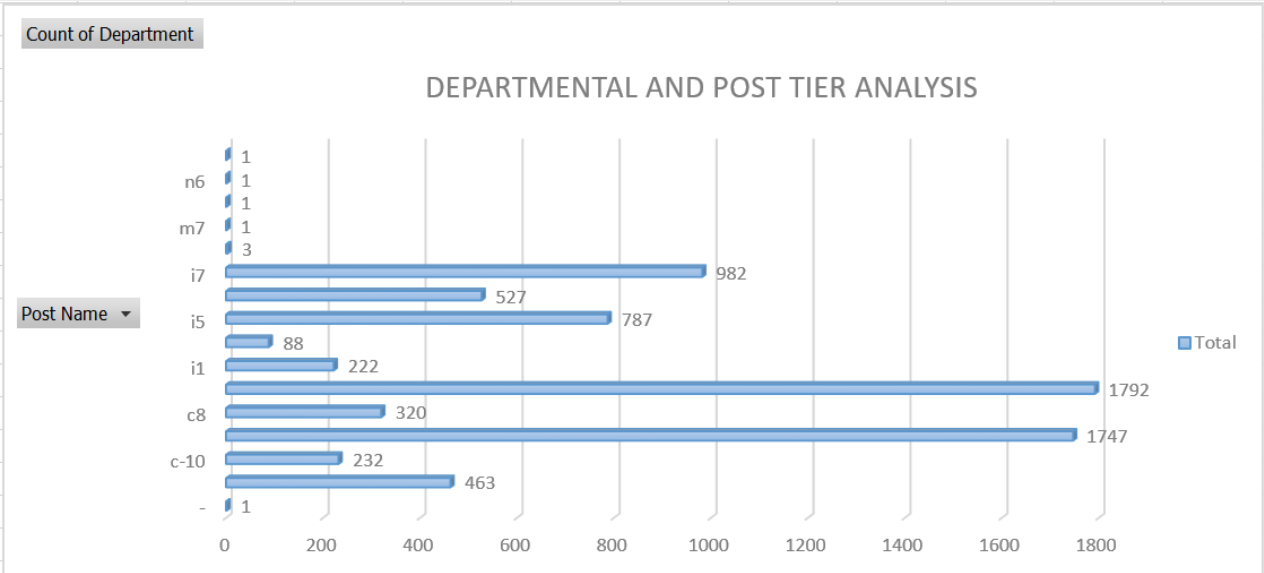


Departmental Analysis: Visualizing data through charts and plots is a crucial part of data analysis.

Your Task: Use a pie chart, bar graph, or any other suitable visualization to show the proportion of people working in different departments.

Position Tier Analysis: Different positions within a company often have different tiers or levels.

Task: Use a chart or graph to represent the different position tiers within the company. This will help you understand the distribution of positions across different tiers.



Row Labels	Count of Department
-	1
b9	463
c-10	232
c5	1747
c8	320
c9	1792
i1	222
i4	88
i5	787
i6	527
i7	982
m6	3
m7	1
n10	1
n6	1
n9	1
Grand Total	7168

RESULT:

- 1) The number of males hired =2563, while the number of females hired =1856.
Males>Females.
- 2) The average salary is **49983(with outliers)**. (outliers excluded) the average salary is 49878.33.

- 3) The majority of people in the dataset have salaries within the range of 100,000.
- 4) A significant portion of the workforce is concentrated in the **operations and service departments**.
- 5) The most common job titles in the dataset are **C9 and C5**.

Hyperlink the Excel Sheet file

https://docs.google.com/spreadsheets/d/1yQkog5OW74vuhJuBdOBuXufivJYsG2-/edit?usp=drive_link&ouid=105315506034365522545&rtpof=true&sd=true

5.IMDB MOVIE ANALYSIS

Project Description:

The objective of this project is to explore the IMDb movie dataset to uncover the key factors that influence a movie's success. Success is measured by high IMDb ratings, and this information is valuable for producers, directors, and investors who want to make data-driven decisions for future projects. This analysis will focus on various aspects like movie genres, duration, languages, directors, and budgets to provide meaningful insights for the entertainment industry.

Approach:

The project involves several phases: data cleaning, analysis, and reporting. First, the dataset is cleaned to handle missing values, ensure data consistency, and remove duplicates. Once the data is prepped, Excel is used for statistical analysis, visualizations, and correlation studies. The 'Five Whys' method is applied to dig deeper into the underlying reasons for each finding. For instance, if we find a correlation between movie budgets and IMDb ratings, we continue asking "why" until we arrive at a meaningful, actionable insight.

Tech-Stack Used:

The primary tool used is **Microsoft Excel 2022**, which was instrumental in performing descriptive statistics, creating visualizations, and calculating correlation coefficients. Its user-friendly interface allows easy manipulation of data and the generation of insights. Additionally, Excel functions like COUNTIF, AVERAGE, and CORREL played a crucial role in the analysis.

Insights

DATA CLEANING:

This step involves preprocessing the data to make it suitable for analysis. It includes handling missing values, removing duplicates, converting data types if necessary, and possibly feature engineering.

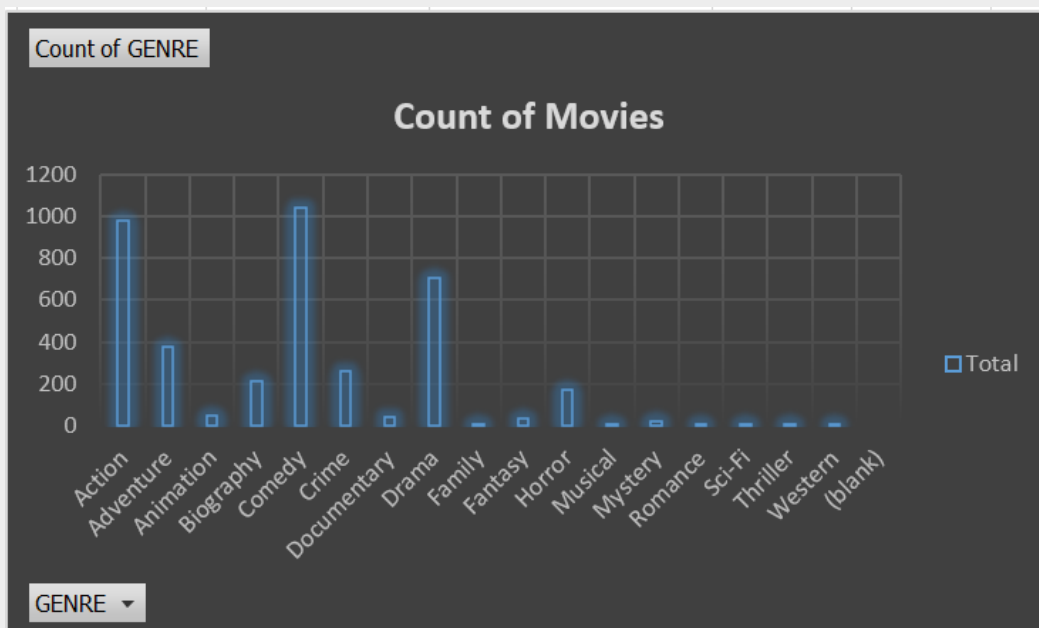
- **Dropping unnecessary columns.**
- **Remove blank space/ Null values.**
- **Remove duplicate**

Movie Genre Analysis:

Analyse the distribution of movie genres and their impact on the IMDB score.

- **Task:** Determine the most common genres of movies in the dataset. Then, for each genre, calculate descriptive statistics (mean, median, mode, range, variance, standard deviation) of the IMDB scores.

GENRE	Count of GENRE	Average	Median	Mode	Variance	SD	Max	Min
Action	976	6.3	6.3	6.6	1.1	1.04	9	2.1
Adventure	380	6.5	6.7	7.3	1.3	1.1	8.6	2.3
Animation	47	6.7	7	7.1	1	1	8	4.5
Biography	209	7.1	7.2	7	0.5	0.7	8.9	4.5
Comedy	1041	6.2	6.3	6.4	1.1	1	8.8	1.9
Crime	262	6.9	7	7.4	0.8	0.9	9.3	3.1
Documentary	43	7.0	7.3	7.3	4	2	8.5	1.6
Drama	704	6.8	6.9	7.3	0.8	0.9	8.8	2.1
Family	3	6.5	5.9	N/A	1.5	1.2	7.9	5.7
Fantasy	37	6.3	6.5	6.8	0.8	0.9	7.9	4.3
Horror	173	5.8	5.8	5.9	1.1	1.1	8.5	2.2
Musical	2	6.8	6.75	N/A	N/A	N/A	7.2	7.2
Mystery	23	6.7	6.7	7.1	1.2	1.1	8.5	3.3
Romance	2	6.7	6.65	N/A	N/A	N/A	7.1	7.1
Sci-Fi	10	6.0	6.2	N/A	1.4	1.2	8.2	5.0
Thriller	3	5.3	4.8	N/A	0.6	0.8	4.5	3.4
Western	4	6.5	6.4	N/A	1.4	2.5	7.3	3.8
(blank)								
Grand Total	3919							

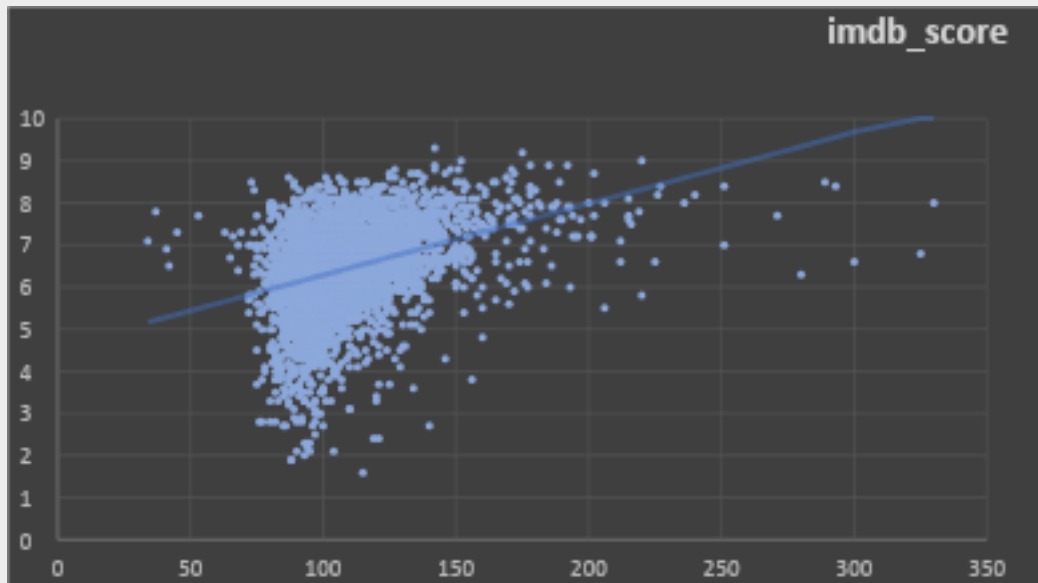


- Used Excel's COUNTIF function to count the number of movies for each genre.
- I had to manipulate the 'genres' column to separate multiple genres for a single movie.
- Used Excel's functions like AVERAGE, MEDIAN, MODE, MAX, MIN, VAR, and STDEV to calculate descriptive statistics.
- Compared the statistics to understand the impact of genre on movie ratings.

CONCLUSION: COMEDY GENRE HAS HIGHEST IMPACT ON IMDB FOLLOWED BY ACTION AND DRAMA.

Movie Duration Analysis: Analyse the distribution of movie durations and its impact on the IMDB score.

- Task: Analyse the distribution of movie durations and identify the relationship between movie duration and IMDB score.



Average	109.78
Median	106.00
STD.	22.69

- Calculated descriptive statistics such as mean, median, and standard deviation for movie durations.
- Used Excel's functions like AVERAGE, MEDIAN, and STDEV. Create a scatter plot to visualize the relationship between movie duration and IMDB score.
- Added a trendline to assess the direction and strength of the relationship.

CONCLUSION:

descriptive analysis	
Mean	6.449859658
Standard Error	0.017050391
Median	6.6
Mode	6.7
Standard Deviation	1.067387146
Sample Variance	1.13931532
Kurtosis	1.090614707
Skewness	-0.731782336
Range	7.7
Minimum	1.6
Maximum	9.3
Sum	25277
Count	3916

Language Analysis: Situation: Examine the distribution of movies based on their language.

- **Task:** Determine the most common languages used in movies and analyze their impact on the IMDB score using descriptive statistics.
- Used Excel's COUNTIF function to count the number of movies for each language.
- Calculated the mean, median, and standard deviation of the IMDB scores for each language.
- Compared the statistics to understand the impact of language on movie ratings.

language	Count of Movies	Average of imdb	Median	s.d
Aboriginal	2	7.2	7.0	0.8
Arabic	1	7.1	7.2	0.0000
Aramaic	1	4.3	7.1	0.0000
Bosnian	1	7.2	4.3	0.0000
Cantonese	8	7.4	7.3	0.44
Czech	1	7.9	7.4	0.0000
Danish	3	7.5	8.1	0.53
Dari	2	7.6	7.5	0.1414
Dutch	3	7.5	7.8	0.4041
Dzongkha	1	6.4	7.5	0.0000
English	3735	6.7	6.5	1.0624
Filipino	1	7.3	6.7	0.0000
French	37	7.7	7.2	0.5613
German	13	7.5	7.7	0.6409
Hebrew	3	6.8	7.3	0.4359
Hindi	10	7.1	7.1	1.1118
Hungarian	1	6.9	7.1	0.0000
Icelandic	1	7.9	6.9	0.0000
Indonesian	2	7.2	7.9	0.4243
Italian	7	7.6	7.0	1.1553
Japanese	12	6.0	7.8	0.8996
Kazakh	1	7.7	6.0	0.0000
Korean	5	7.1	7.7	0.5701
Mandarin	15	7.8	7.4	0.7720
Maya	1	7.3	7.8	0.0000
Mongolian	1	8.5	7.3	0.0000
None	1	7.2	8.5	0.0000
Norwegian	4	8.1	7.3	0.5745
Persian	3	7.8	8.4	0.5508
Portuguese	5	7.9	8.0	0.9788
Romanian	1	6.5	7.9	0.0000
Russian	1	7.1	6.5	0.0000
Spanish	26	7.6	7.2	0.8262
Swedish	1	8.4	7.6	0.0000
Telugu	1	6.6	8.4	0.0000
Thai	3	7.4	6.6	0.4509
Vietnamese	1	7.3	7.4	0.0000
Zulu	1	7.1	7.3	1.0000
(blank)				
Grand Total	3916			

CONCLUSION:

ENGLISH WITH 3735.

Director Analysis: Influence of directors on movie ratings.

- Task: Identify the top directors based on their average IMDB score and analyze their contribution to the success of movies using percentile calculations.

Top 10 Directors	Average of imdb_score
Charles Chaplin	8.60
Tony Kaye	8.60
Alfred Hitchcock	8.50
Damien Chazelle	8.50
Majid Majidi	8.50
Ron Fricke	8.50
Sergio Leone	8.43
Christopher Nolan	8.43
Asghar Farhadi	8.40
Marius A. Markevicius	8.40



- Calculated the average IMDB score for each director.
- Used Excel's PERCENTILE function to identify the directors with the highest scores.
- Compared the scores of these directors to the overall distribution of scores.

CONCLUSION:

HIGHEST	8.60
PERCENTILE RANK	0.89
PERCENTILE	8.60

Budget Analysis: Explore the relationship between movie budgets and their financial success.

- Task: Analyze the correlation between movie budgets and gross earnings, and identify the movies with the highest profit margin.

- Calculated the correlation coefficient between movie budgets and gross earnings using Excel's CORREL function.
- Calculated the profit margin (gross earnings - budget) for each movie and identify the movies with the highest profit margin using Excel's MAX function.

CONCLUSION

Hight profit margin movie	HIGHT PROFIT MARGIN
The Hobbit: The Desolation of Smaug	523505847

CORRELATION
0.102177361

EXCEL SHEET LINK

https://docs.google.com/spreadsheets/d/1voYzzK7EVhqvRyrtaa5q_qsE494mPntj/edit?usp=drive_link&ouid=105315506034365522545&rtpof=true&sd=true

6.BANK LOAN CASE STUDY

As a data analyst at an urban-focused finance company, my mission is clear: navigate the complexities of loan approvals to ensure that only capable, reliable customers are granted loans. My role isn't merely about numbers and patterns; it's about safeguarding the company's financial health while making fair lending decisions. The challenges are significant—many applicants don't have extensive credit histories, and some may exploit this gap, resulting in loan defaults. But by thoroughly analysing loan application data, I can bring clarity to decision-making processes, separating the responsible applicants from the risky ones.

The Challenge

In this context, two primary risks define every loan decision:

1. **Missed Opportunity:** If an applicant with the ability to repay is rejected, the company loses valuable business.
2. **Financial Loss:** Conversely, if a customer who is likely to default is approved, the company risks a financial setback.

The data provides essential insights into customer patterns, allowing us to distinguish between two main types of cases:

- **Payment Difficulties:** Customers who were late by more than a certain number of days on at least one of their early payments.

- **On-Time Payments:** Customers who managed to pay consistently without significant delays.

Every application falls into one of four outcomes:

- **Approved:** The customer's loan request was accepted.
- **Cancelled:** The applicant cancelled the application midway.
- **Refused:** The application was rejected.
- **Unused Offer:** Although approved, the applicant chose not to use the loan.

My goal is to identify how different customer and loan characteristics correlate with repayment behaviour, helping to make more effective lending decisions.

Target & Goal

TARGET	GOAL
• To use Explorator Analysis (EDA) to find patterns in the data	
• To identify pattern indicate if a customer difficulty paying the instalments.	• To ensure that capable applicants are not rejected.
• To understand the reason behind loan default	• To make decisions such as denying the loan, reducing the amount of loan, or lending at a higher interest rate to risky applicants
	• To make better decisions about loan approval.

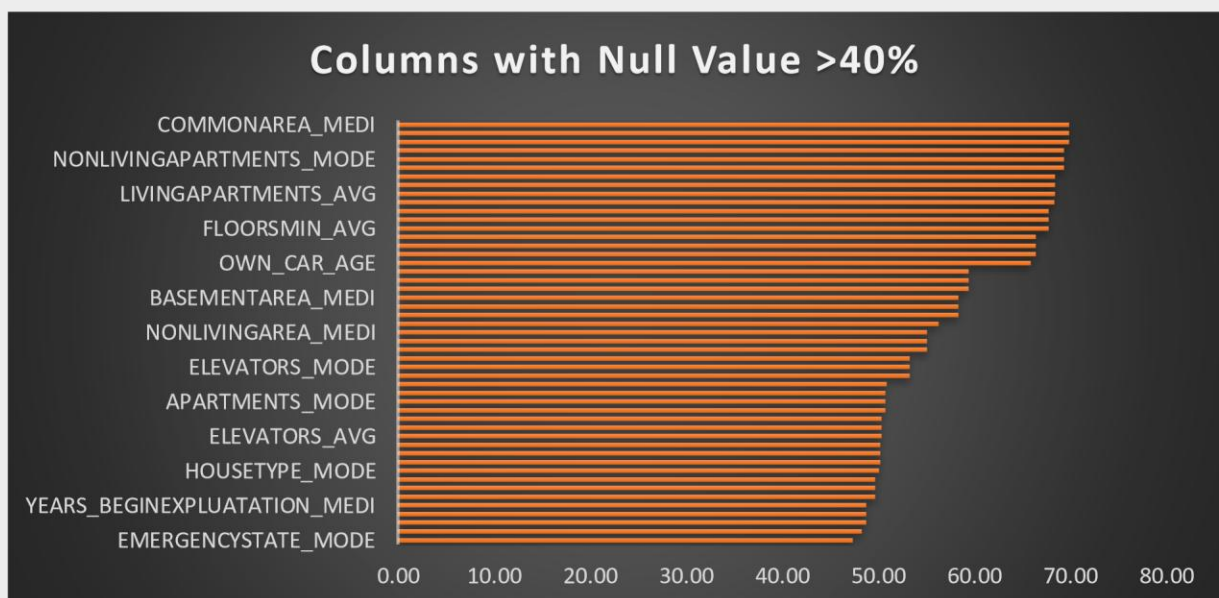
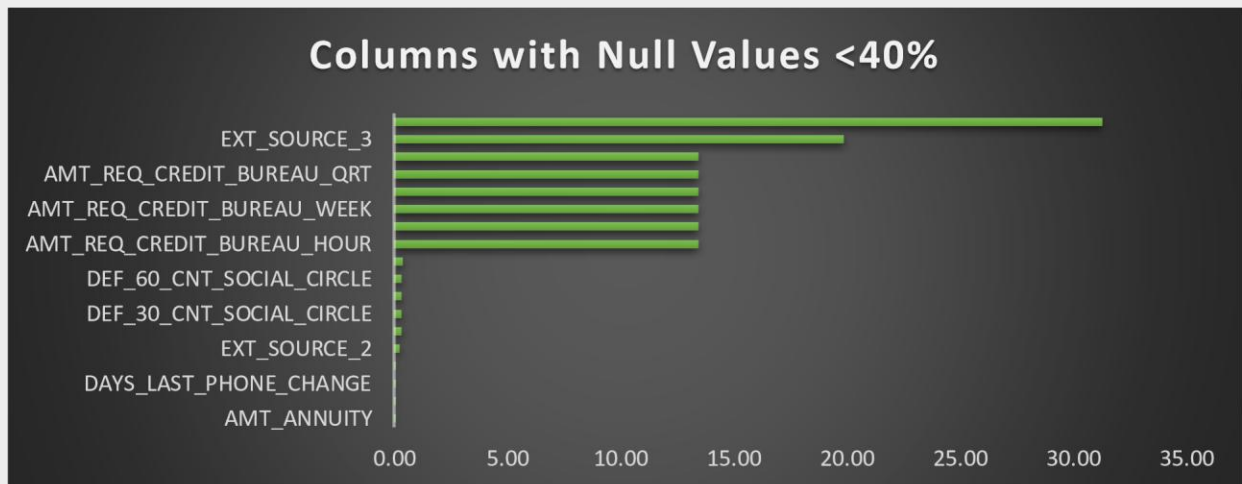
DATA ANALYTICS TASKS TO BE PERFORMED

IDENTIFYING MISSING DATA

The missing data in the dataset and decide on an appropriate method to deal with it using

Excel built-in functions and features.

- Select relevant columns from both data sets which will be used for further analysis.
- Fill the missing data with the “=MEDIAN” of that column to prevent uneven scattering of data.
- Use “=COUNTBLANK(range)” to count number of missing values in each column.
- Store this data in different table.
- Create “Bar Chart” from this table to visualize better on number of missing values in data.

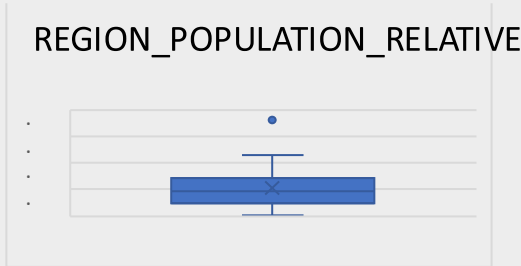
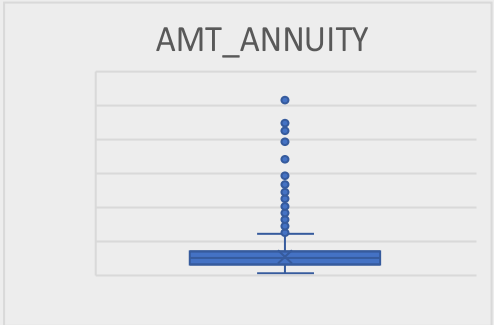
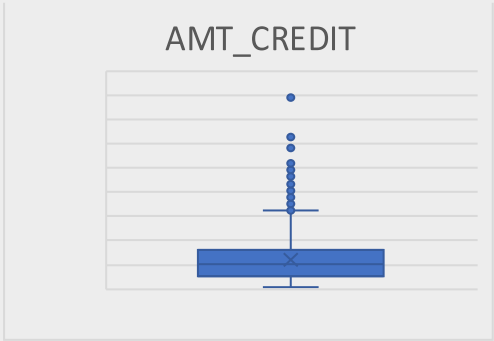


IDENTIFY OUTLIERS IN THE DATASET

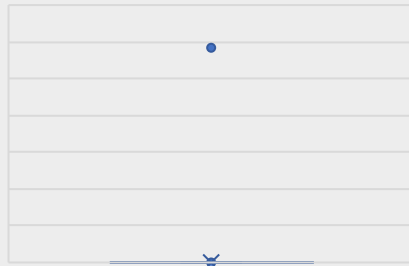
Detect and identify outliers in the dataset using Excel statistical functions and features, focusing on numerical variables.

- Use “=QUARTILE(range,1)” to find Q1.
- Use “=QUARTILE(range,3)” to find Q3.
- Find **Interquartile Range (IQR)** by subtracting Q1 from Q3.
- Find lower limit of threshold range by using “Q1- 1.5*IQR”.
- Find upper limit of threshold range by using “Q3+1.5*IQR”.
- Apply “**CONDITIONAL FORMATTING**” to find values greater than the upper limit of threshold.
- Create “**BOX & WHISKER PLOT**” for each column to find outliers of that column.

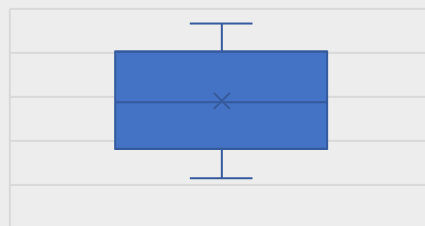
	Q1	Q3	IQR	Upper Limit	Lower Limit
CNT_CHILDREN	0	1	1	2.5	-1.5
AMT_CREDIT	270000	808650	538650	1616625	-537975
AMT_ANNUITY	16456.5	34596	18139.5	61805.25	-10752.75
AMT_GOODS_PRICE	238500	679500	441000	1341000	-423000
REGION_POPULATION_RELATIVE	0.01001	0.02866	0.01866	0.0566485	-0.0179795
Client_Age	33.9123	53.8192	19.9068	83.6794521	4.0520548
Yrs_Employed	2.55616	15.6658	13.1096	35.330137	-17.108219
Yrs_Registration	5.47397	20.4479	14.974	42.9089041	-16.986986



AMT_INCOME_TOTAL



AMT_GOODS_PRICE

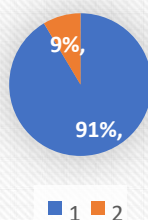


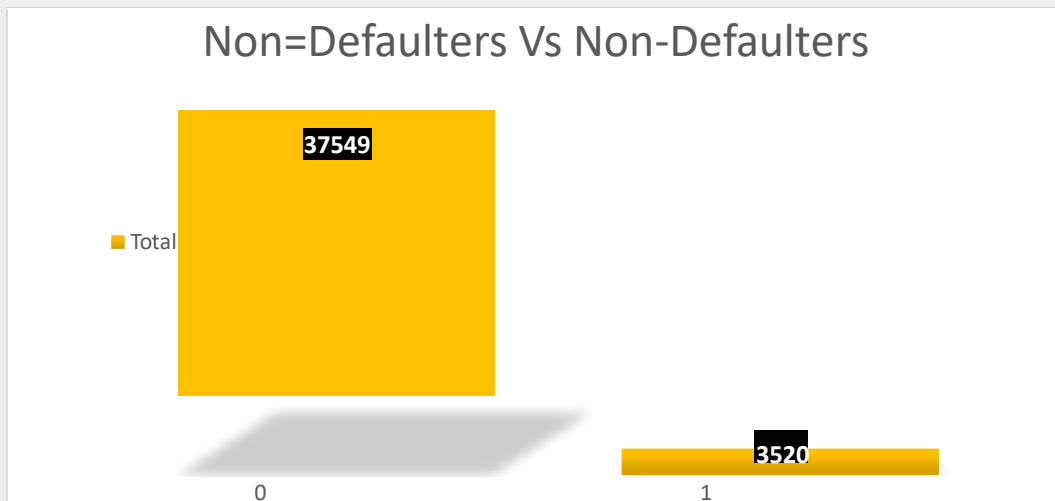
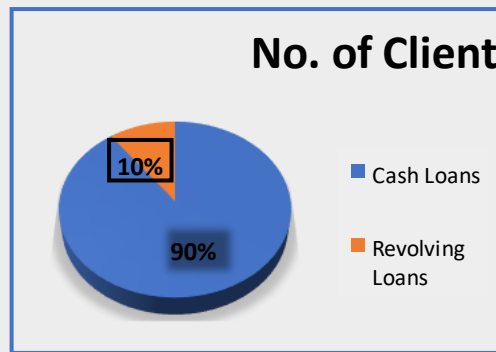
ANALYSE DATA IMBALANCE

Determine if there is data imbalance in the loan application dataset and calculate the ratio of data imbalance using Excel functions.

- Use “**=COUNTIF(range,criteria)**” to find number of class-0 and class-1 category in TARGET column.
- To find CLASS PROPORTION for each class divide count of that class by total number of data.
- Find **IMBALANCE RATIO** by dividing lower number of class counts by greater number of class count.
- Create **PIE CHART** to better visualize the imbalance of data.

Ratio of target



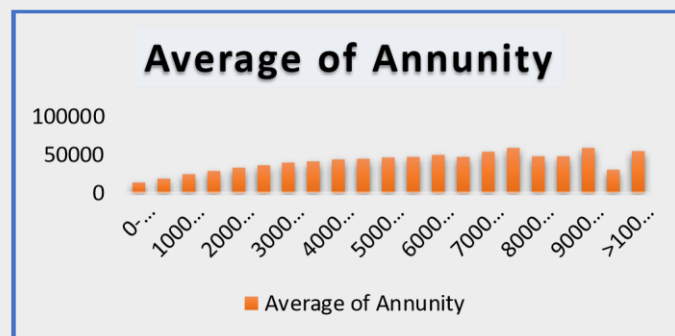
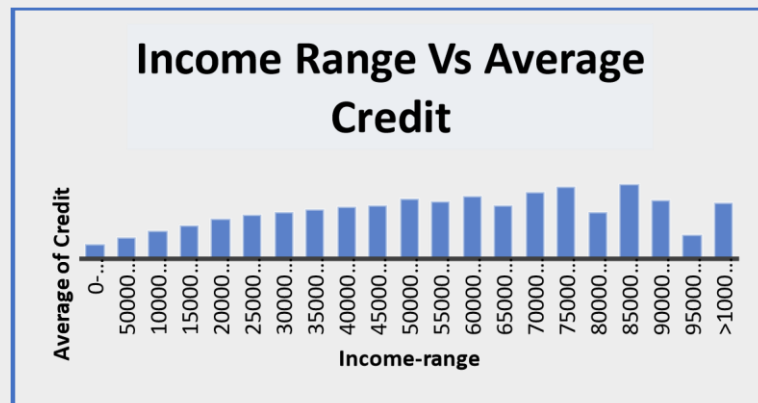


There are a greater number of “No payment difficulty” type of customers.

PERFORM UNIVARIATE, SEGMENTED UNIVARIATE, AND BIVARIATE ANALYSIS

Perform-

- univariate analysis to understand the distribution of individual variables.
- segmented univariate analysis to compare variable distributions for different scenarios.
- bivariate analysis to explore relationships between variables and the target variable using Excel functions
- Segregate required tables.
- For **UNIVARIATE**, apply “**BOX & WHISKER**” method or “**COLUMN GRAPH**” for analysis.
- Apply “**PIVOT TABLE**” on 2 or more columns for **BIVARIATE** Analysis.
- Create “**STACK COLUMN**” and “**COLUMN GRAPH**” for analysis.



IDENTIFY TOP CORRELATIONS FOR DIFFERENT SCENARIOS

Segment the dataset based on different scenarios (e.g., clients with payment difficulties and all other cases) and identify the top correlations for each segmented data using Excel functions.

- Select the columns whose co-relation is to be found.
- Use “=CORREL(range1, range2)” function to find the co-relation factor.

Top correlation of Non-Defaulters			
Rank	Variable 1	Variable 2	Correlation
1	AMT_GOODS_PRICE	AMT_CREDIT	0.98635817
2	REGION_RATING_CLIENT_W_CITY	REGION_RATING_CLIENT	0.95028652
3	CNT_CHILDREN	CNT_FAM_MEMBERS	0.8937356
4	REG_REGION_NOT_WORK_REGION	LIVE_REGION_NOT_WORK_REGION	0.8601677
5	DEF_30_CNT_SOCIAL_CIRCLE	DEF_60_CNT_SOCIAL_CIRCLE	0.85304075
6	REG_CITY_NOT_WORK_CITY	LIVE_CITY_NOT_WORK_CITY	0.81560498
7	REGION_RATING_CLIENT	AMT_GOODS_PRICE	0.76520174
8	AMT_ANNUITY	AMT_GOODS_PRICE	0.76520174
9	AMT_CREDIT	AMT_ANNUITY	0.76082787

Top Correlation of Defaulters			
Rank	Variable 1	Variable 2	Correlation
1	AMT_GOODS_PRICE	AMT_CREDIT	0.98192814
2	REGION_RATING_CLIENT	REGION_RATING_CLIENT_W_CITY	0.94802081
3	CNT_FAM_MEMBERS	CNT_CHILDREN	0.89560034
4	DEF_60_CNT_SOCIAL_CIRCLE	DEF_30_CNT_SOCIAL_CIRCLE	0.89146724
5	LIVE_REGION_NOT_WORK_REGION	REG_REGION_NOT_WORK_REGION	0.80558323
6	LIVE_CITY_NOT_WORK_CITY	REG_CITY_NOT_WORK_CITY	0.77310735
7	AMT_ANNUITY	AMT_GOODS_PRICE	0.74642245
8	AMT_ANNUITY	AMT_CREDIT	0.74513211

PROJECT INSIGHTS

- The following conclusions can be made from this project-
- Missing values must be filled with MEDIAN value of that column to prevent uneven scattering of the data.
- There are no missing values in "AMT_APPLICATION" and "AMT_CREDIT" from Previous_Application dataset.
- "AMT_DOWN_PAYMENT" column from Previous_Application dataset has largest number of missing values.
- "AMT_GOODS_PRICE" from Application_Dataset has largest number of missing values.
- Maximum number of columns in Application_Dataset has minimum or no missing values.
- There are many outliers in maximum number of columns.
- "AMT_INCOME_TOTAL" and "DAYS_EMPLOYED" from Application_Dataset has only 1 outlier.
- Maximum number of loans have no difficulty in payments.
- 4026 loans have difficulties in payment.
- Imbalance ratio is of 0.09.
- Consumer loans are most common type of loans.
- Maximum loans are approved.
- Very few offers are left unused.
- Total income and annuity amount are less dependent than annuity amount and credit amount.

[https://docs.google.com/spreadsheets/d/19Qlw3gTd27NecLb4JLNoG-mXdU0L9Ts/edit?usp=drive link&ouid=105315506034365522545&rtpof=true&sd=true](https://docs.google.com/spreadsheets/d/19Qlw3gTd27NecLb4JLNoG-mXdU0L9Ts/edit?usp=drive_link&ouid=105315506034365522545&rtpof=true&sd=true)

7.“ANALYSING THE IMPACT OF CAR FEATURES ON PRICE AND PROFITABILITY”

PROJECT DESCRIPTION:

This project aims to help a car manufacturer optimize pricing and product development decisions to maximize profitability while meeting consumer demand. By analyzing key car features like engine power, fuel type, and market category, this project seeks to identify which attributes most significantly impact pricing, popularity, and profitability.

Business Problem:

With rising competition and consumer preferences shifting towards fuel-efficient and technologically advanced vehicles, manufacturers need data-driven insights to understand consumer demand and develop competitive pricing strategies.

Data Source:

The dataset, "Car Features and MSRP," contains 11,159 observations and 16 variables and was originally sourced from Kaggle. It includes details on car models, specifications, and pricing information up to the year 2017.

Data Cleaning and Preprocessing Steps:

1. **Handle Missing Values:** Check for and impute any missing data for essential variables.
2. **Data Type Conversion:** Ensure all numeric fields (e.g., MSRP, horsepower) are in a consistent numeric format.
3. **Standardization of Categorical Variables:** Confirm consistency in category names (e.g., fuel types, transmission types).
4. **Remove Duplicates:** Ensure no duplicate entries for unique car models.

Approach

Analytical Methods Used:

- **Descriptive Statistics:** To provide a summary overview of the variables and their distributions.

- **Visualization Techniques:** Utilize pivot tables, scatter plots, combo charts, and bar charts to explore relationships between variables.
- **Regression Analysis:** Determine the key features that influence MSRP and their relative importance.
- **Market Segmentation:** Analyse different market categories for trends in pricing and popularity.

Reasoning for Method Selection: The analysis methods were chosen to uncover both direct and nuanced relationships between car features and price, which are essential for deriving insights for product development and pricing strategies.

Tech Stack Used

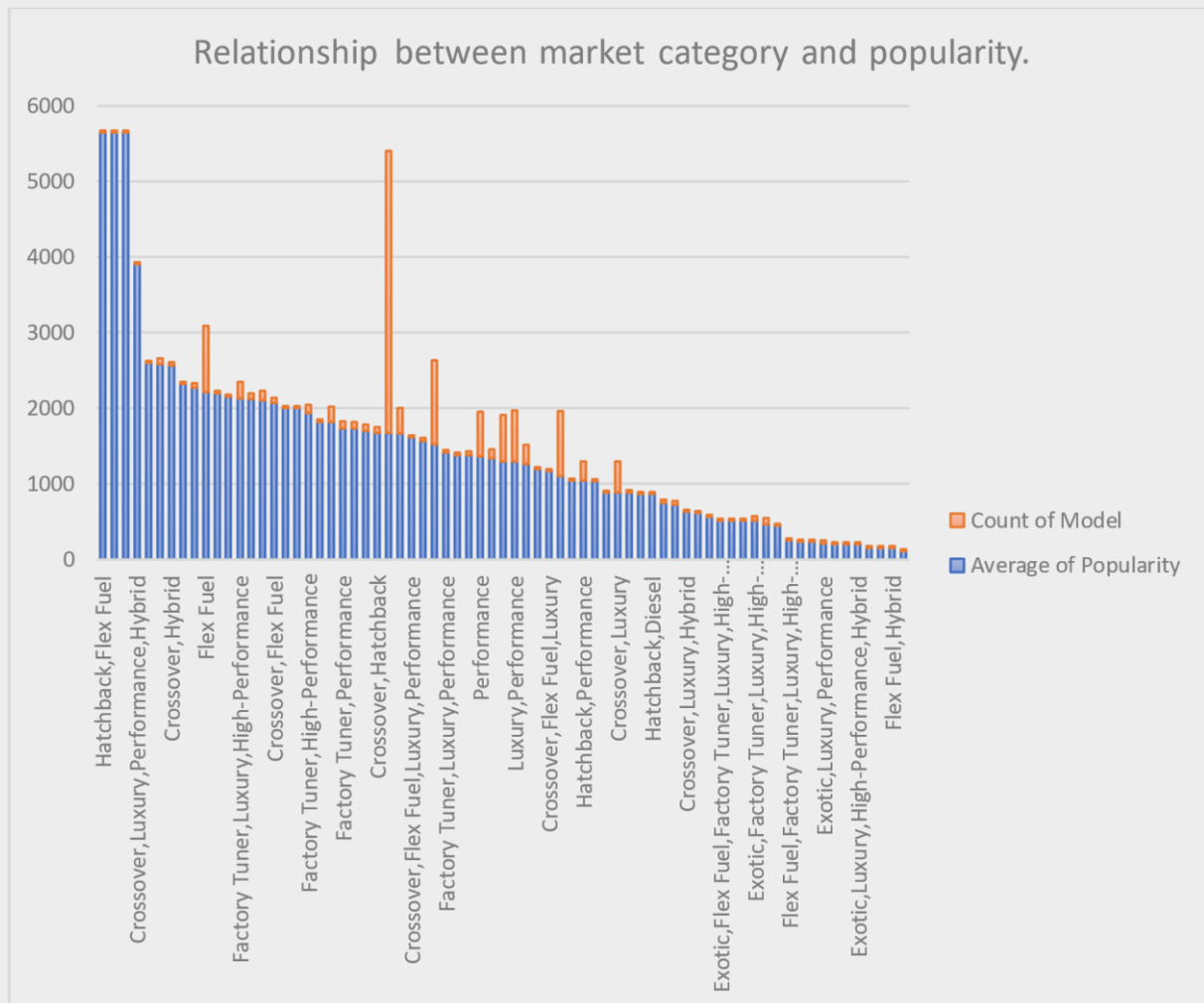
- **Tool:** Microsoft Excel, chosen for its capabilities in creating pivot tables, interactive dashboards, and performing statistical analysis with regression.
- **Additional Resources:** Slicers, filters, and combo charts for interactivity; AVERAGEIF, SUMIF, and similar functions for calculations.

INSIGHTS

Task 1

How does the popularity of a car model vary across different market categories?

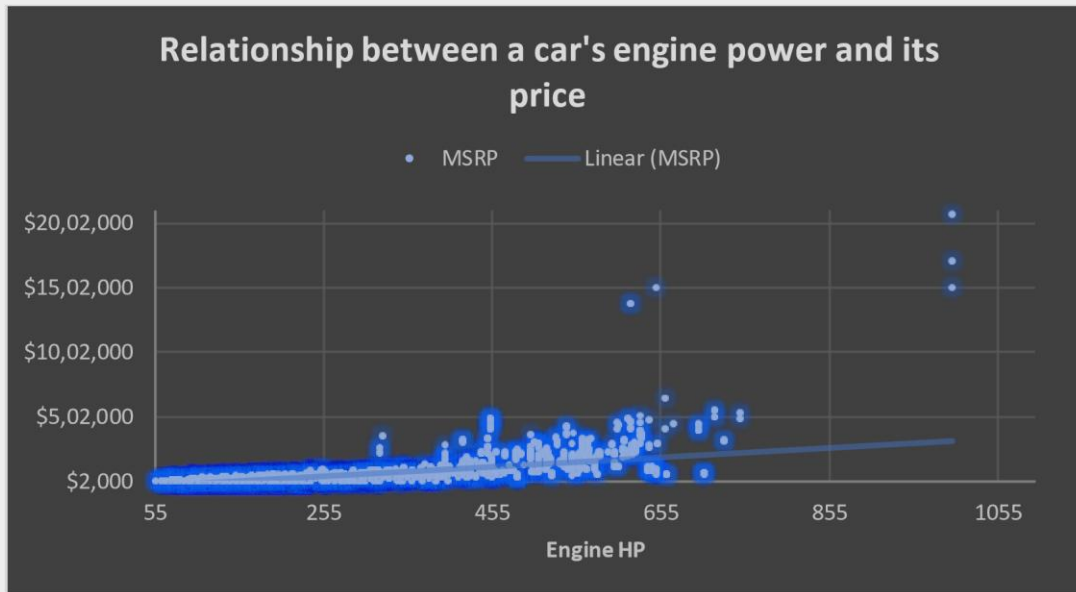
- *Use "INSERT> PIVOT TABLE" command on the market category and popularity score columns.*
- *Find "SUM" of each category score through it.*
- *Create a "COLUMN CHART" for the same.*



Task 2 What is the relationship between a car's engine power and its price?

- Select “ENGINE HP” and “MSRP” columns.
- The apply the following command to get the relationship between the two-

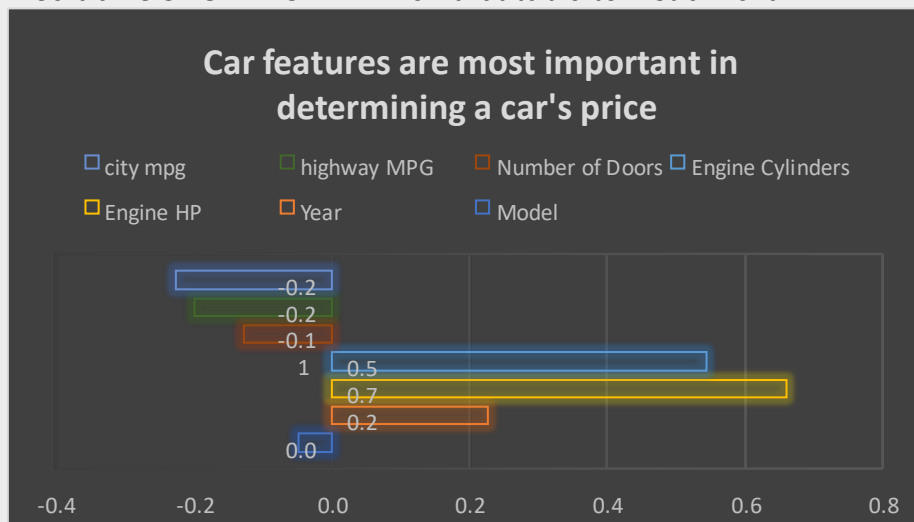
INSERT> CHART> SCATTER PLOT> ADD TRENDLINE



Task 3

Which car features are most important in determining a car's price?

- Use “=CORREL(array1,array2)” function on “MSRP” and every other column category whose data is of integer type.
- Store the correlation factors of different categories with respect to MSRP in another table.
- Insert a “COLUMN CHART” on that table to visualize it.



	Model	Year	Engine HP	Engine Cylinders	Number of Doors	highway MPG	city mpg
MSRP	0.0	0.2	0.7	0.5	-0.1	-0.2	-0.2

Regression analysis to identify the variables that have the strongest relationship with a car's price.

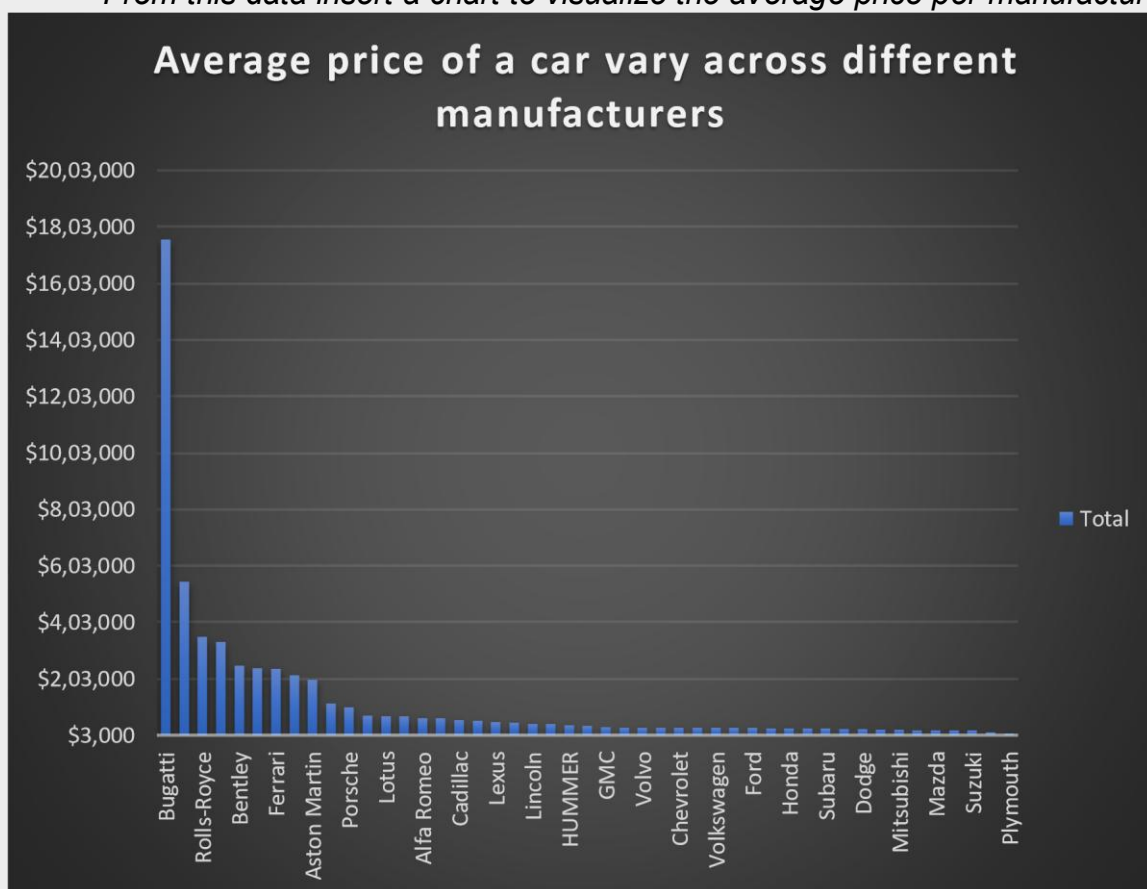
Task 4

How does the average price of a car vary across different manufacturers?

- Select “MANUFACTURER” and “MSRP” column and run the following command-

“INSERT> PIVOT TABLE”

- Find average price using the “PIVOT TABLE”.
- From this data insert a chart to visualize the average price per manufacturer.



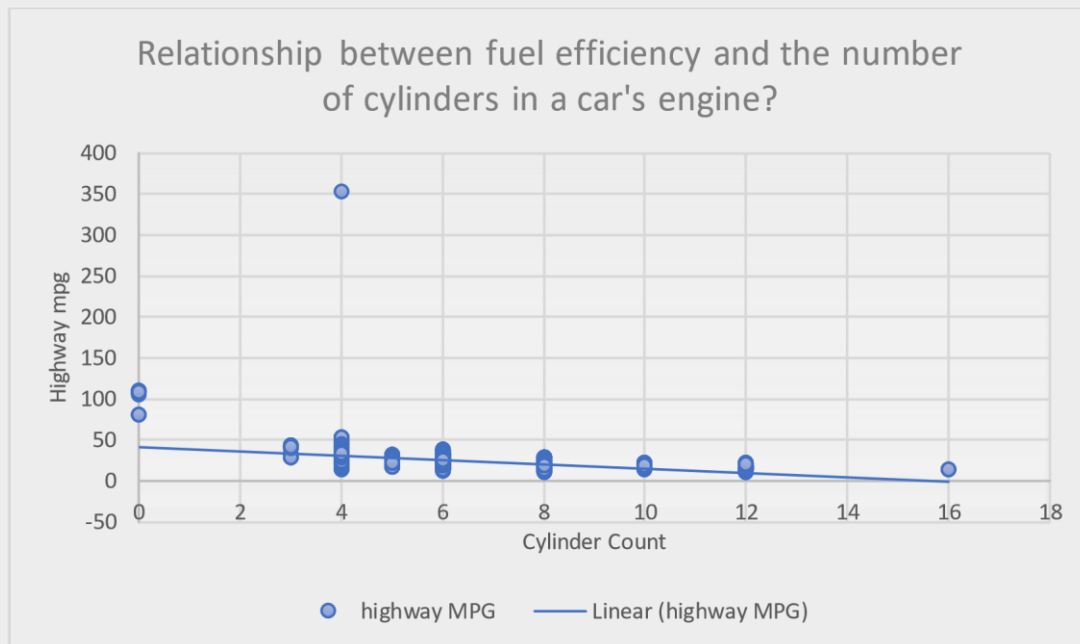
Task 5

What is the relationship between fuel efficiency and the number of cylinders in a car's engine?

- Select “NO. OF CYLINDERS” and “HIGHWAY MPG” column and run the following command-

“INSERT> CHART> SCATTER PLOT>ADD TRENDLINE”

- Use “=CORREL(array1,array2)” command on same columns to find the correlation factor and direction of correlation as well.



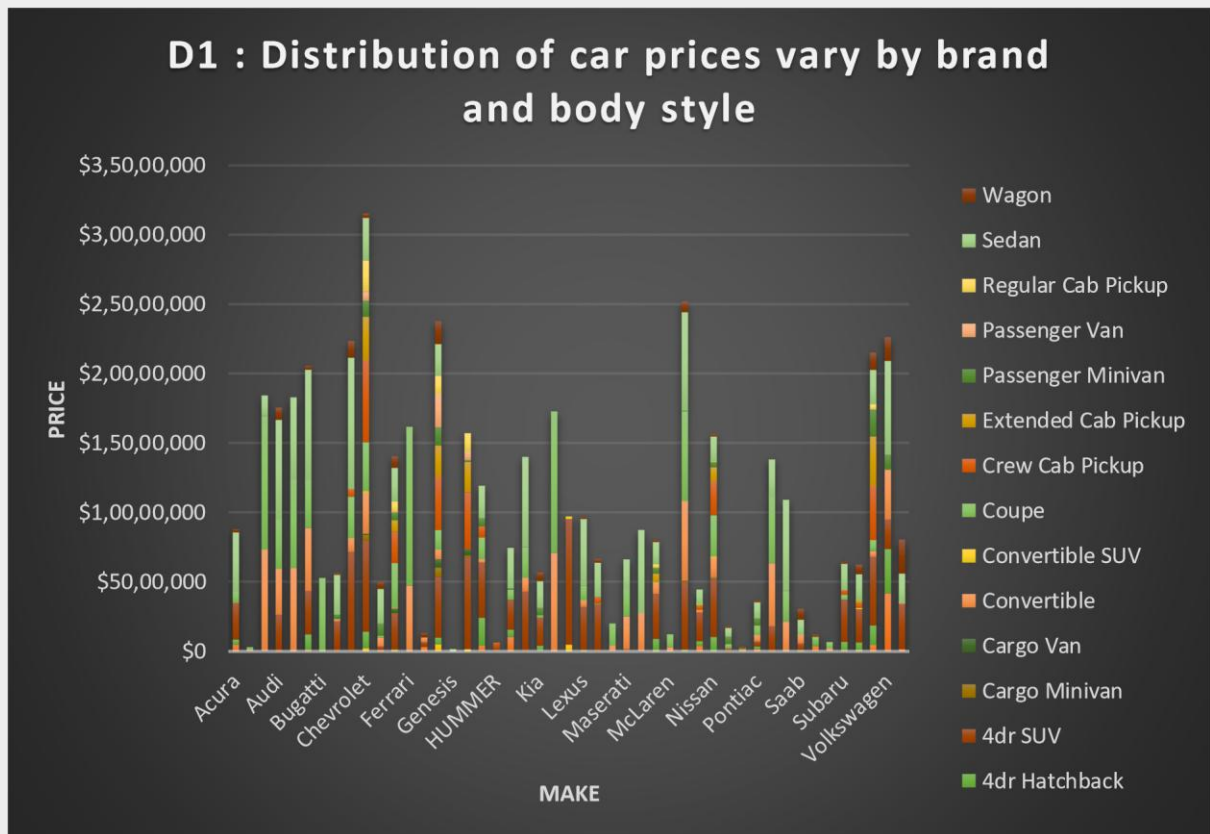
Dashboard Task 1

How does the distribution of car prices vary by brand and body style?

- Select “MSRP”, “VEHICLE STYLE” and “MAKE” column and run the following command-

“INSERT> PIVOT TABLE

- Use “SUM” for value field setting.
- Create a stack column chart from pivot table created.



Dashboard Task 2

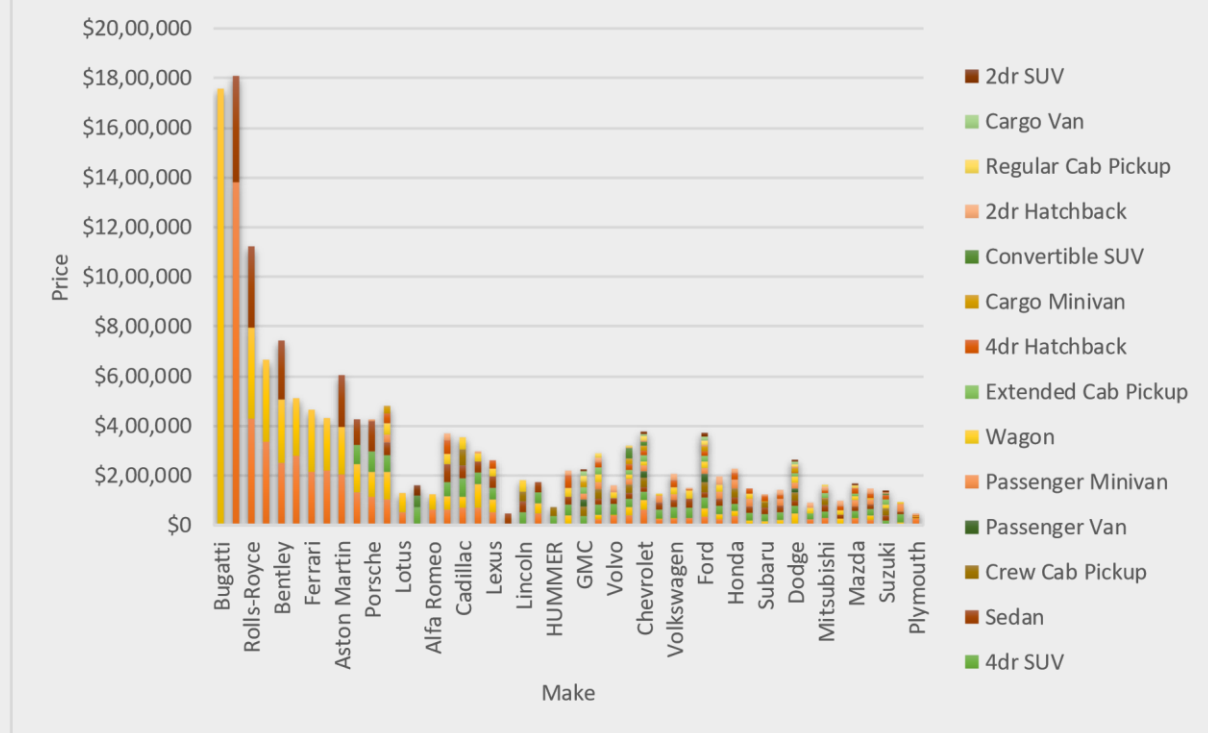
Which car brands have the highest and lowest average MSRPs, and how does this vary by body style?

- Select “MSRP”, “VEHICLE STYLE” and “MAKE” column and run the following command-

“INSERT> PIVOT TABLE

- Use “AVG” for value field setting.
- Create a clustered column chart from pivot table created.

D2 : Car brands have the highest and lowest average MSRPs, and This is how it vary by body style



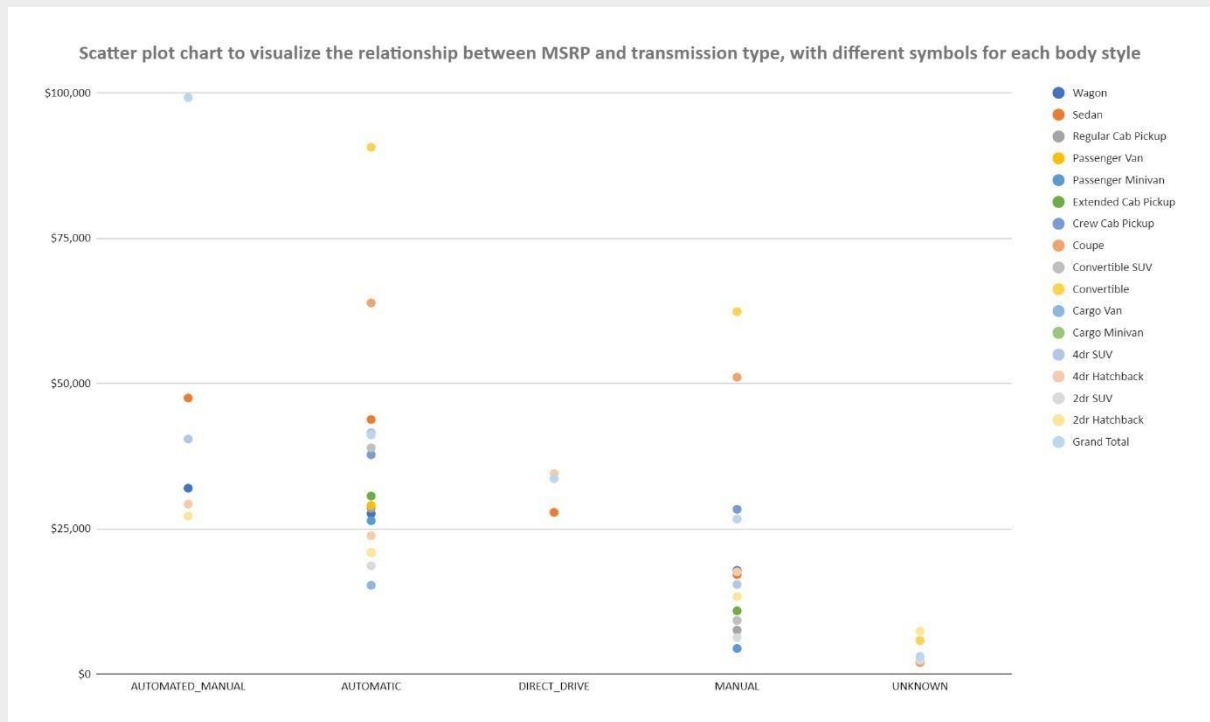
Dashboard Task 3

How do the different feature such as transmission type affect the MSRP, and how does this vary by body style?

- Select "MSRP", "VEHICLE STYLE" and "TRANSMISSION TYPE" column and run the following command-

"INSERT> PIVOT TABLE

- Use "AVG" for value field setting.
- Create a stack column chart from pivot table created.



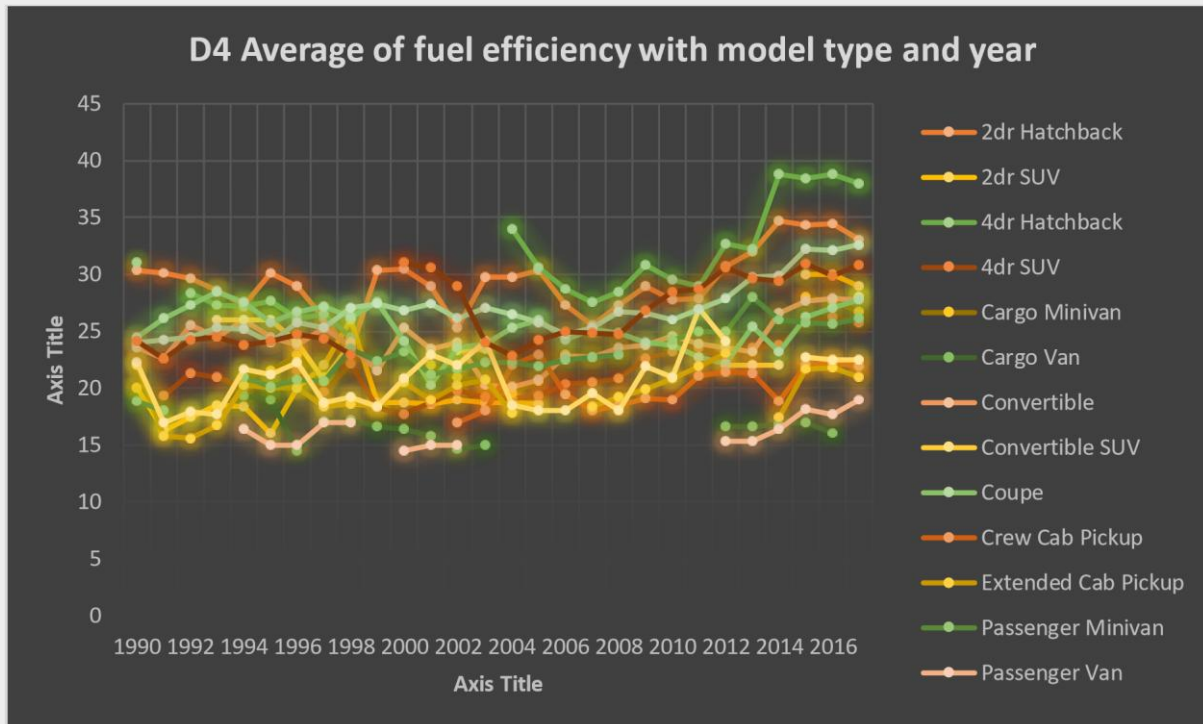
Dashboard Task 4

How does the fuel efficiency of cars vary across different body styles and model years?

- Select “HIGHWAY MPG” and “VEHICLE STYLE” column and run the following command-

“INSERT> PIVOT TABLE

- Use “AVG” for value field setting.
- Create a stack column chart and line chart from pivot table created.



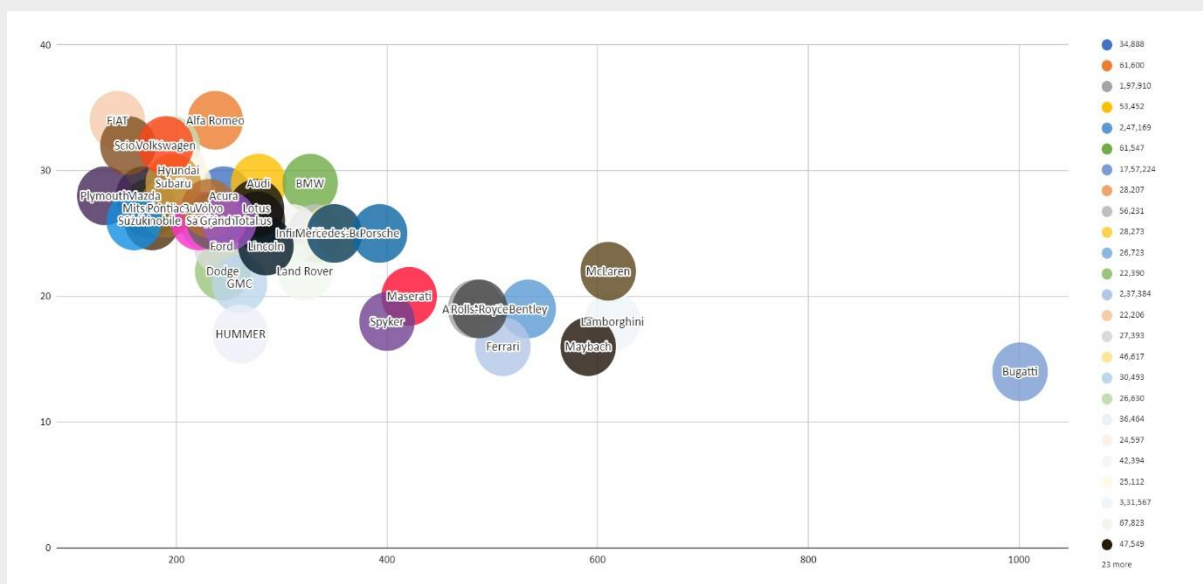
Dashboard Task 5

How does the car's horsepower, MPG, and price vary across different Brands?

- Select “ENGINE HP”, “MSRP” and “HIGHWAY MPG” column and run the following command-

“INSERT> PIVOT TABLE

- Use “AVG” for value field setting.
- Create a stack column chart and line chart from pivot table created.



Results

Visualization:

To make the findings clear and actionable, we used visual tools like pivot tables, scatter plots, and bar charts. These visuals simplify complex relationships in the data, enabling stakeholders to quickly grasp the key insights and make informed decisions.

Discussion:

The analysis highlights essential trends for car manufacturers, such as consumer preferences for specific car features and how these impact pricing and popularity. Understanding these dynamics allows manufacturers to tailor their strategies, align with consumer demand, and increase profitability in a competitive market.

Future Directions:

In the future, further analysis could explore adaptive pricing strategies, integrate realtime market trends, and use advanced modelling for more precise forecasts. Regularly tracking consumer preferences and industry changes will be essential to staying relevant and competitive.

Conclusion:

The insights gained from this analysis provide valuable guidance for car manufacturers to refine pricing and product development strategies, ensuring they remain aligned with a rapidly evolving market.

LINK FOR EXCEL FILE:

https://docs.google.com/spreadsheets/d/1cZJhri1plsA9V-RuAqJ53XuXX5DIlzg0/edit?usp=drive_link&oid=105315506034365522545&rtpof=true&sd=true

8.ABC Call Volume Trend Analysis

Project Description

This project focuses on Customer Experience (CX) analytics, specifically analysing the inbound calling team's data over 23 days. The dataset includes details such as agent names and IDs, queue times (how long customers wait to connect), call times, durations, and statuses (answered, abandoned, or transferred).

CX teams play a vital role by analysing customer feedback, gathering data insights, and sharing these with the broader organization. With the support of modern tools like Interactive Voice Response (IVR) and Predictive Analytics, CX teams work to enhance customer satisfaction and improve overall service.

Inbound support agents handle customer inquiries, aiming to engage and satisfy customers, transforming them into loyal advocates. In addition to CX analysis, this project considers the impact of advertising, as first impressions shape customer views. With many companies competing for the same audience, our analysis will uncover patterns in call volume, helping the company optimize its audience targeting efficiently.

TECH STACK USED

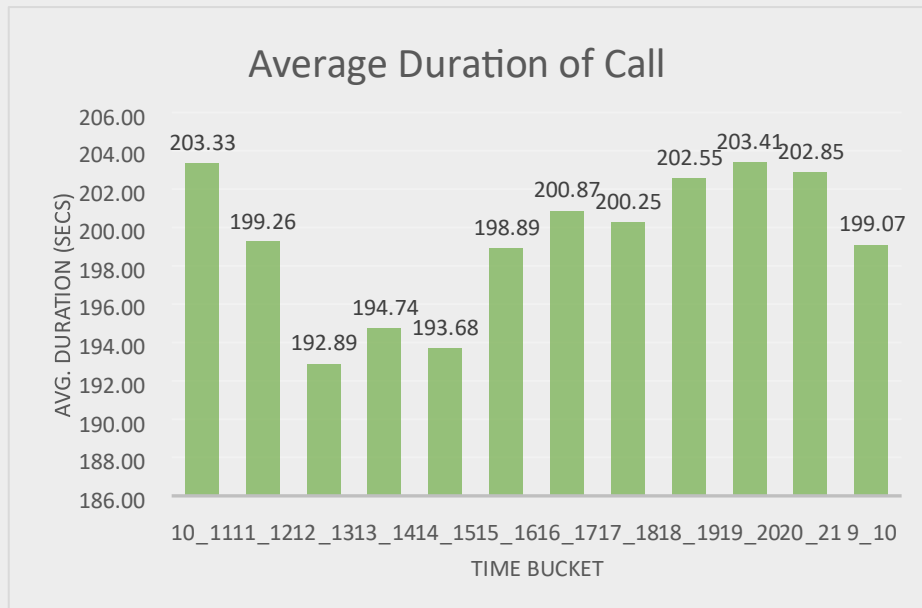
MS EXCEL: It has user-friendly interface, versatile functions, and calculation capabilities make it essential for data management, financial analysis, and business reporting.

Data Analytics Tasks

Average Call Duration

Determine the average duration of all incoming calls received by agents. This should be calculated for each time bucket.

Your Task: What is the average duration of calls for each time bucket?

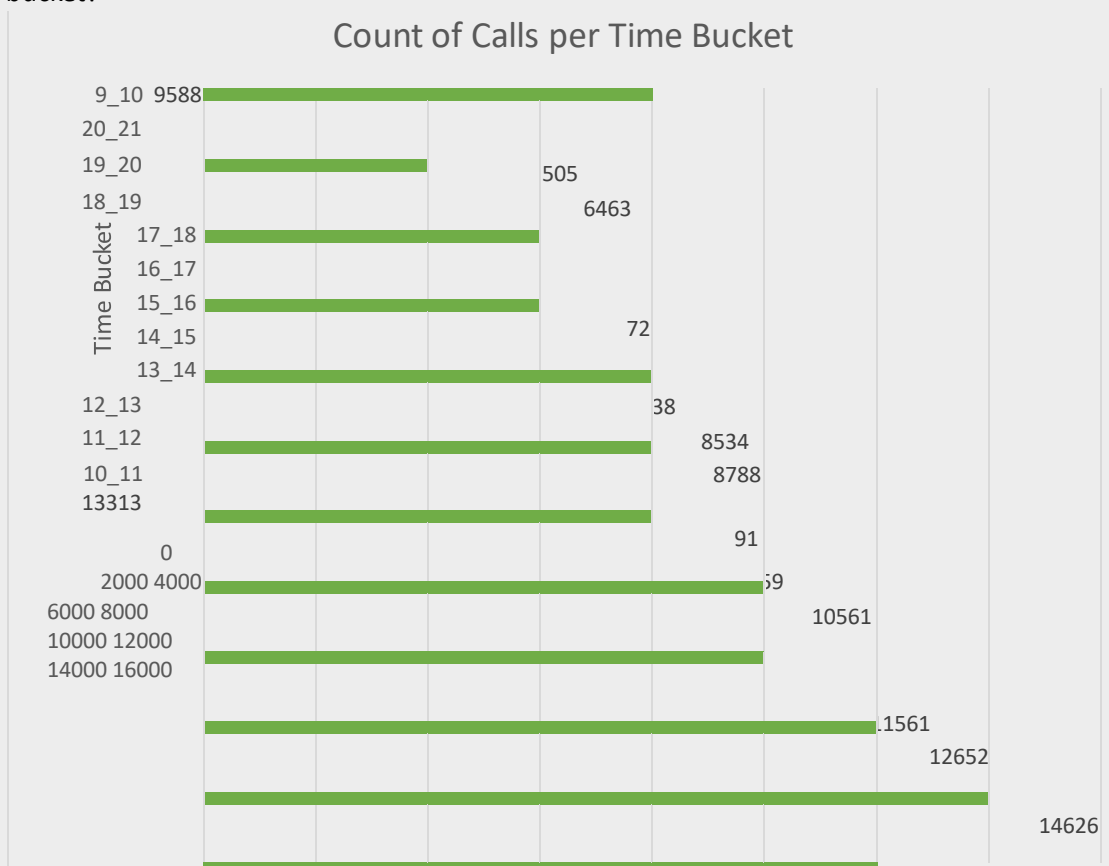


The highest average and lowest average is in 10-11 and 12-13 respectively.

Call Volume Analysis

Visualize the total number of calls received. This should be represented as a graph or chart showing the number of calls against time. Time should be represented in buckets (e.g., 1-2, 2-3, etc.).

Your Task: Can you create a chart or graph that shows the number of calls received in each time bucket?



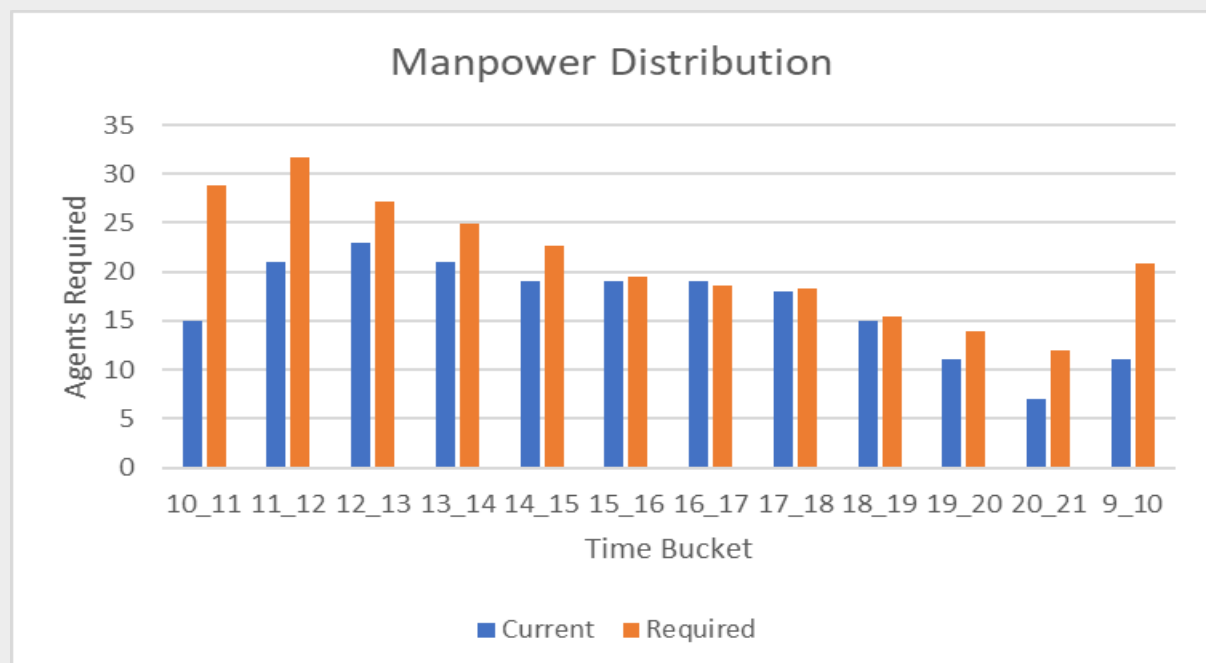
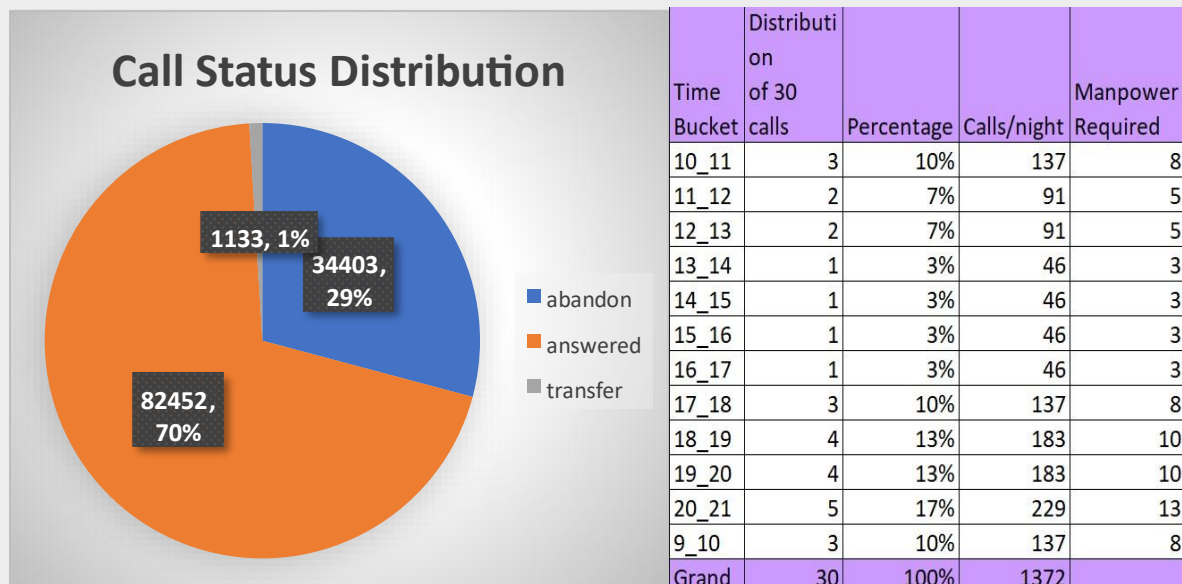
Number of Calls

11-12 is the MOST BUSIEST TIME SLOT and 20_21 is THE LEAST BUSIEST TIME SLOT.

Manpower Planning

The current rate of abandoned calls is approximately 30%. Propose a plan for manpower allocation during each time bucket (from 9 am to 9 pm) to reduce the abandon rate to 10%. In other words, you need to calculate the minimum number of agents required in each time bucket to ensure that at least 90 out of 100 calls are answered.

Your Task: What is the minimum number of agents required in each time bucket to reduce the abandon rate to 10%?



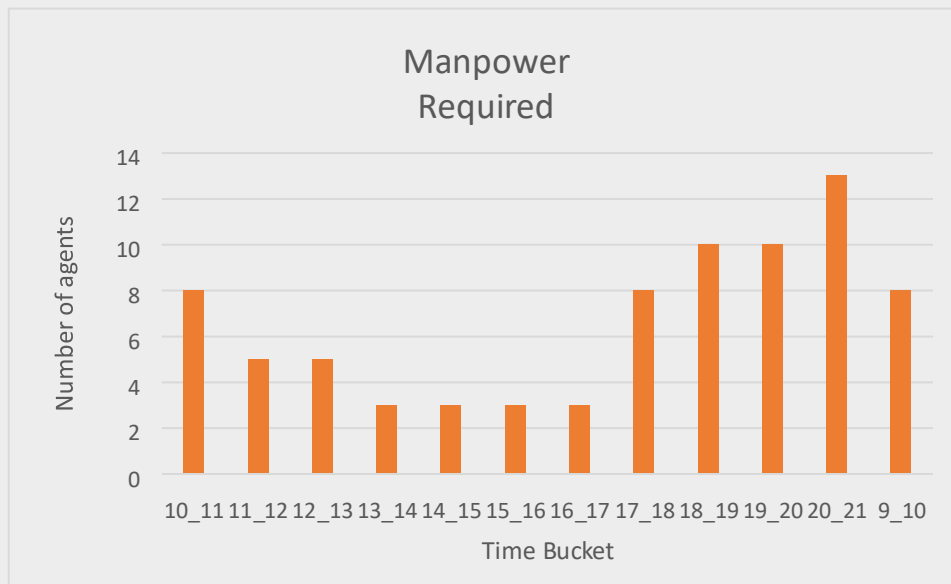
The time slot of 11-12 requires the MAXIMUM NUMBER OF AGENTS, while the slot of 12-13 holds the maximum number of agents current.

Night Shift Manpower Planning

Customers also call ABC Insurance Company at night but don't get an answer because there are no agents available. This creates a poor customer experience. Assume that for every 100 calls that customers make between 9 am and 9 pm, they also make 30 calls at night between 9 pm and 9 am. The distribution of these 30 calls is as follows:

Your Task: Propose a manpower plan for each time bucket throughout the day, keeping the maximum abandon rate at 10%.

Time Bucket	Distribution of 30 calls	Percentage	Calls/night	Manpower Required
10_11	3	10%	137	8
11_12	2	7%	91	5
12_13	2	7%	91	5
13_14	1	3%	46	3
14_15	1	3%	46	3
15_16	1	3%	46	3
16_17	1	3%	46	3
17_18	3	10%	137	8
18_19	4	13%	183	10
19_20	4	13%	183	10
20_21	5	17%	229	13
9_10	3	10%	137	8
Grand	30	100%	1372	



The time bucket of 20-21 holds maximum of the agents with a count of 13, while the slots of 13-14,14-15,14-15 and 15-16 are the lowest and almost equally holding number of agents.

HYPERLINK TO EXCEL SHEET:

https://docs.google.com/spreadsheets/d/1MZ5nAHWkDoJha4BTXhe7xIDS73KDXKzC/edit?usp=drive_link&ouid=105315506034365522545&rtpof=true&sd=true

THANKYOU.